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Nordlander's Theorem
–On the modulus of convexity–

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Chapter 1

Introduction to Banach spaces

In this chapter we introduce some basic concepts and instruments in the theory of Banach spaces. All the notions are standard, and can be found for example in [Fol99].

1.1 Normed Linear Spaces

Definition 1.1. Let X be a linear space. A real-valued function $\|\cdot\|$ on X is called a **norm** provided for each f and g in X and each real number α:

i $\|f+g\| \leqslant \|f\| + \|g\|$;

ii $\|\alpha f\| = |\alpha|\|f\|$;

iii $\|f\| \geqslant 0$ *and* $\|f\| = 0$ *if and only if* $f = 0$.

By a **normed linear space** we mean a linear space together with a norm $\|\cdot\|$.

Definition 1.2. A sequence $\{f_n\}_{n\in\mathbb{N}}$ in a linear space X that is normed by $\|\cdot\|$ is said to **converge** to f provided that:

$$\lim_{n\to\infty} \|f_n - f\| = 0.$$

Definition 1.3. A sequence $\{f_n\}_{n\in\mathbb{N}}$ in a linear space X that is normed by $\|\cdot\|$ is said to be **Cauchy** in X provided that for each $\varepsilon > 0$ there exist $n_0 \in \mathbb{N}$ such that

$$\|f_n - f_m\| < \varepsilon \ \forall m,n \geqslant n_0.$$

remark 1.4. All **convergent** sequences in a normed linear space X are **Cauchy**, but the opposite in not generally true.

Definition 1.5. A normed linear space is said to be **complete** if and only if all Cauchy sequences converge in X. A complete normed linear space is called a **Banach space**.

In this section, we present an equivalent characterization of Banach spaces. Intermediary lemma's are required before proving the main result.

Lemma 1.6. *Let $(X, \|\cdot\|)$ be a normed linear space. Let $\{x_n\}_{n\in\mathbb{N}}$ be a Cauchy sequence. If there exists a subsequence $\{x_{n_k}\}_{k\in\mathbb{N}} \subset \{x_n\}_{n\in\mathbb{N}}$ that converges, then $\{x_n\}_{n\in\mathbb{N}}$ converges.*

Proof. Consider a Cauchy sequence $\{x_n\}_{n\in\mathbb{N}} \subset X$ with a convergent subsequence $\{x_{n_k}\}_{k\in\mathbb{N}} \subset \{x_n\}_{n\in\mathbb{N}}$. Let x be said limit.
Let $\varepsilon > 0 \implies \exists\, N$ s.t. $\|x_n - x_m\| < \varepsilon\ \forall n, m \geqslant N$. Let $k \in \mathbb{N}$ s.t. $\|x_{n_k} - x\| < \varepsilon$ and $n_k \geqslant N$. Then $\|x_n - x\| \leqslant \|x_n - x_{n_k}\| + \|x_{n_k} + x\| < 2 \cdot \varepsilon$ QED

Lemma 1.7. *Let $(X, \|\cdot\|)$ be a normed linear space. $\{x_n\}_{n\in\mathbb{N}}$ a Cauchy sequence $\implies \exists\, \{x_{n_k}\}_{k\in\mathbb{N}} \subset \{x_n\}_{n\in\mathbb{N}}$ s.t. $\|x_{n_{k+1}} - x_{n_k}\| \leqslant 2^{-k}\ \forall k \in \mathbb{N}$.*

Proof. Let $k = 0,\ \varepsilon = 2^0 :\ \exists\, n_0$ s.t. $\|x_n - x_m\| < 2^0\ \forall n, m \geqslant n_0$. For $k = 1,\ \varepsilon = 2^{-1} :$ $\exists\, n_1 > n_0$ s.t. $\|x_n - x_m\| < 2^{-1}\ \forall n, m \geqslant n_1$. In general let $\varepsilon = 2^{-k} :\ \exists\, n_k > n_{k-1}$ s.t. $\|x_n - x_m\| < 2^{-k}\ \forall n, m \geqslant n_k$. For $\varepsilon = 2^{-(k+1)} :\ \exists\, n_{k+1} > n_k$ s.t. $\|x_n - x_m\| < 2^{-(k+1)}\ \forall n, m \geqslant n_{k+1} \implies \|x_{n_{k+1}} - x_{n_k}\| < 2^{-k}$ for $n_{k+1} > n_k$. QED

Proposition 1.8. *Let $(X, \|\cdot\|)$ be a normed linear space. $(X, \|\cdot\|)$ is complete $\iff$ $\sum_{n\in\mathbb{N}} \|x_n\| \lneqq \infty \implies \sum_{n\in\mathbb{N}} x_n$ converges.*

A standard reference is [Fol99].

Proof. If X is complete and $\sum_{n=1}^{\infty} \|x_n\| < \infty$, let $S_N = \sum_{n=1}^{N} x_n$. Then for $N > M$ we have

$$\|S_N - S_M\| \leqslant \sum_{n=M+1}^{N} \|x_n\| \xrightarrow[M,N\to\infty]{} 0.$$

So the sequence $\{S_N\}$ is Cauchy. Hence convergent.
Conversely, suppose that every absolutely convergent series converges. Let $\{x_n\}$ be a Cauchy sequence. Lemma 1.7 allow us to choose $n_1 < n_2 < \dots$ such that $\|x_m - x_k\| < 2^{-k+1}$ for $m, k \geqslant n_k$. Let $y_1 = x_{n_1}$ and $y_j = x_{n_j} - x_{n_{j-1}}$ for $j > 1$. Then $\sum_{j=1}^{k} y_j = x_{n_k}$, and

$$\sum_{j=1}^{\infty} \|y_j\| \leqslant \|y_1\| + \sum_{j=2}^{\infty} 2^{-j+1} = \|y_1\| + 1 < \infty.$$

So $\lim_{k\to\infty} x_{n_k} = \sum_{j=1}^{\infty} y_j$ exists. Since $\{x_n\}$ is Cauchy, lemma 1.6 proves that $\{x_n\}$ converges to the same limit as $\{x_{n_k}\}$. QED

1.2 L^p spaces are Banach spaces

Here we introduce an important class of Banach spaces: L^p spaces for $p \geqslant 1$. We will prove that the set of all measurable functions is a complete normed linear space when paired with the norm $\|\cdot\|_{L^p}$.

Consider a measure space $(\Omega,\ \mathscr{M},\ \mu)$ with μ non negative.

Consider a measurable function $f : \Omega \longrightarrow \mathbb{R}$.

Definition 1.9. Let $f, g : \Omega \longrightarrow \mathbb{R}$ be measurable functions.

$f \sim g \Longleftrightarrow \mu\left(\{x \in \Omega : f(x) \neq g(x)\}\right) = 0.$

remark 1.10. $\sim$ is a equivalence relationship.

From this point forward let f be a representative of an equivalence class of $\sim$.

Definition 1.11. Let $f : X \longrightarrow \mathbb{C}$ be a measurable function.
For $p \in [1, \infty)$: $\|f\|_p = \left(\int |f|^p \, d\mu\right)^{\frac{1}{p}}$;

$$\|f\|_\infty = \begin{cases} \|f\|_\infty = +\infty \text{ if } \{\alpha :\ |f| \leqslant \alpha \ a.e.\} = \emptyset \\ \|f\|_\infty = \inf\{\alpha :\ |f| \leqslant \alpha \ a.e.\} \text{ if } \{\alpha :\ |f| \leqslant \alpha \ a.e.\} \neq \emptyset \end{cases}$$

where $\{\alpha :\ |f| \leqslant \alpha \ a.e.\} = \{\alpha \geqslant 0 :\ \mu(\{x \in \Omega :\ |f(x)| \geqslant \alpha\}) = 0\}$.

Definition 1.12. $L^p(\Omega,\ \mu) = \{f : \Omega \longrightarrow \mathbb{C} \ \textit{measurable} :\ \|f\|_p \lneqq \infty\}$

remark 1.13. $L^p := L^p(\Omega,\ \mu)$ is real vector space.

To prove that $\|\cdot\|_p$ is a norm, we'll need the following proposition:

Lemma 1.14. *If $a \geqslant 0, b \geqslant 0$, and $0 \gneqq \lambda \gneqq 1$, then: $a^\lambda \cdot b^{1-\lambda} \leqslant \lambda \cdot a + (1-\lambda) \cdot b$*

Proof. The result is obvious if $b = 0$; otherwise, dividing both sides by b and setting $t = a/b$, we are reduced to showing that

$$t^\lambda \leqslant \lambda t + (1-\lambda)$$

with equality $\Longleftrightarrow t = 1$. $t^\lambda - \lambda t$ is strictly increasing for $t < 1$ and strictly decreasing for $t > 1$, so its maximum value, namely $1 - \lambda$, occurs at $t = 1$. QED

Proposition 1.15. *Let f, g be measurable functions on X, then*

$$\|f \cdot g\| \leqslant \|f\|_p \cdot \|g\|_q \ \textit{for} \ \frac{1}{p} + \frac{1}{q} = 1,\ p \in [1, \infty].$$

Proof. The result is satisfied if $\|f\|_p = 0$ or $\|g\|_q = 0$ (since then $f = 0$ or $g = 0$ a.e.), or if $\|f\|_p = \infty$ or $\|g\|_q = \infty$. For $p = \infty,\ q = 1$ we have that $\int |f \cdot g| \leqslant \int \|f\|_\infty \cdot |g| = \|f\|_\infty \cdot \|g\|_1$. Let $p \neq 1, \infty$ and assume WLOG $\|f\|_p = \|g\|_q = 1$. Applying lemma 1.14 with $a = |f(x)|^p$, $b = |g(x)|^q$ and $\lambda = \frac{1}{p}$, we obtain:

$$|f(x)g(x)| \leq \frac{1}{p}|f(x)|^p + \frac{1}{q}|g(x)|^q. \tag{1.16}$$

Integration of both sides yields

$$\|fg\|_1 \leq \frac{1}{p}\int |f|^p + \frac{1}{q}\int |g|^q = \frac{1}{p} + \frac{1}{q} = 1 = \|f\|_p \|g\|_q.$$

QED

Proposition 1.17. *Let f, g be measurable functions on X, then*

$$\|f+g\|_p \leqslant \|f\|_p + \|g\|_p \ p \in [1, \infty].$$

Proof. For $p = 1$, $\|f+g\|_1 = \int |f+g| \leqslant \int |f| + |g| = \int |f| + \int |g| = \|f\|_1 + \|g\|_1$.

For $p = \infty$: note that if $|f(x)| + |g(x)| \leqslant \alpha$ a.e., then

$$|f(x) + g(x)| \leqslant |f(x)| + |g(x)| \leqslant \alpha \ a.e.$$

$$\begin{aligned}
\|f+g\|_\infty \ &= \ \inf\{\alpha \geqslant 0 : \ \mu(\{x \in \Omega : \ |f(x) + g(x)| \geqslant \alpha\}) = 0\} \\
&\leqslant \\
\inf\{\alpha + \beta, \ \alpha, \beta \geqslant 0 : \ \mu(\{x \in \Omega : \ |f(x)| + |g(x)| \geqslant \alpha + \beta\}) &= 0\} \\
&\leqslant \\
\inf\{\alpha \geqslant 0 : \ \mu(\{x \in \Omega : \ |f(x)| \geqslant \alpha\}) &= 0\} \\
+ \ \inf\{\beta \geqslant 0 : \ \mu(\{x \in \Omega : \ |g(x)| \geqslant \beta\}) &= 0\} = \|f\|_\infty + \|g\|_\infty.
\end{aligned}$$

Let $p \neq 1, \infty : (f+g)^p = (f+g)(f+g)^{p-1}$

$$\implies \|f+g\|_p^p = \int |f+g|^p \leqslant \int |f| \cdot |f+g|^{p-1} + \int |f+g|^p \leqslant \int |g| \cdot |f+g|^{p-1}.$$

Applying proposition 1.15:

$$\begin{aligned}
\|f+g\|_p^p &\leqslant \|f\|_p \cdot \||f+g|^{p-1}\|_q + \|g\|_p \cdot \||f+g|^{p-1}\|_q \\
&= \\
&(\|f\|_p + \|g\|_p) \cdot \||f+g|^{p-1}\|_q.
\end{aligned}$$

Note that $p - 1 = \frac{p}{q}$:

$$\implies \||f+g|^{p-1}\|_q = \left(\int |f+g|^{(p-1)\cdot q}\right)^{\frac{1}{q}} = \left(\left(\int |f+g|^p\right)^{\frac{1}{p}}\right)^{\frac{p}{q}} = \|f+g\|_p^{p-1}$$

$$\implies \frac{\|f+g\|_p^p}{\|f+g\|_p^{p-1}} \leqslant \|f\|_p + \|g\|_p.$$

QED

$\|\cdot\|_p$ is a norm and $L^p := L^p(\Omega,\ \mu)$ is a normed linear space.

remark 1.18. Let $f, g : \Omega \longrightarrow \mathbb{R}$ be measurable function, then:

1. $|f(x)| \leqslant \|f\|_\infty$ a.e.;
2. $|f(x)| \leqslant M\ \forall x \in \Omega \implies \|f\|_\infty \leqslant M$ and $f \in L^\infty$;
3. $|f(x)| \leqslant |g(x)|$ a.e. $\implies \|f\|_\infty \leqslant \|g\|_\infty$.

We deduce from point 1 of this remark that

$$\{\alpha :\ |f| \leqslant \alpha\ a.e.\} \neq \emptyset \implies \|f\|_\infty = \min\{\alpha \geqslant 0 :\ \mu\{x \in \Omega :\ |f(x)| \geqslant \alpha\} = 0\}.$$

Proposition 1.19. *Consider the space of bounded measurable functions on Ω defined as*

$$B(\Omega) = \left\{ f : X \longrightarrow \mathbb{C}\ s.t.\ \|f\|_B = \sup_{x\in\Omega} |f(x)| \lneqq \infty \right\}.$$

Then $(B(\Omega), \|\cdot\|_B)$ is a Banach space.

Proof. Let $\{f_n\} \subset B(\Omega)$ be a Cauchy sequence.

$$\forall x \in \Omega : |f_n(x) - f_m(x)| \leqslant \|f_n - f_m\|_B \xrightarrow[n,m\to\infty]{} 0$$

$$\implies \{f_n(x)\} \text{ is a Cauchy sequence in } \mathbb{C}\ \forall x \in \Omega.$$

$\mathbb{C}$ is complete $\implies \{f_n(x)\}$ converges $\forall x \in \Omega$. Call $f(x) := \lim_{n\to\infty} f_n(x)\ \forall x \in \Omega$. We will prove that $f \in B(\Omega)$ and $\|f_n - f\|_B \xrightarrow[n\to\infty]{} 0$. $\forall \varepsilon > 0,\ \exists\, N \in \mathbb{N}$ s.t.

$$\|f_n - f_m\|_B < \varepsilon \implies ||f_n(x) - f_m(x)| - |f_n(x) - f(x)|| \leqslant |f_m(x) - f(x)| \xrightarrow[m\to\infty]{} 0$$

$$\implies \forall n \geqslant N\ |f_n(x) - f(x)| = lim_{m\to\infty} |f_n(x) - f_m(x)| \leqslant \varepsilon$$

$$\implies \|f_n - f\|_B \leqslant \varepsilon\ \forall n \geqslant N \implies \|f_n - f\|_B \xrightarrow[n\to\infty]{} 0.$$

To prove $f \in B(\Omega)$: $\|f\|_B \leqslant \|f_N - f\|_B + \|f_N\|_B \lneqq \infty$. QED

Proposition 1.20. *L^∞ is a Banach space.*

Proof. Let $\{f_n\}_n$ be a Cauchy sequence in L^∞, we want to show that there $\exists\, f \in L^\infty$ s.t. $\|f_n - f\|_\infty \to 0$ as $n \to \infty$. We will do the proof in three steps:

1. find a possible candidate for f;
2. show that $f \in L^\infty$;
3. show that $\|f_n - f\|_\infty \to 0$.

$\forall n \in \mathbb{N}$ consider f_n, and define:

$$E_{m,n} := \{x \in \Omega : |f_m(x) - f_n(x)| > \|f_m - f_n\|_\infty\} \implies \mu(E_{m,n}) = 0,$$

$$A_k := \{x \in \Omega : |f_k(x)| > \|f_k\|_\infty\} \implies \mu(A_k) = 0.$$

Since$\{f_n\}$ are measurable, then $E_{m,n}$ e A_k are measurable and

$$\mu\left(\bigcup_{m,n} E_{m,n} \cup \bigcup_k A_k\right) = 0.$$

Define $E = \left(\bigcup_{m,n} E_{m,n} \cup \bigcup_k A_k\right)^c \implies E$ is measurable and $\mu(E^c) = 0$. Define $g_n = f_n|_E\ \forall n$, and $g_n \equiv 0$ otherwise. Then:

- $g_n \in B(E)\ \forall n$ as $\forall x \in E\ |g_n(x)| = |f_n(x)| \leqslant \|f_n\|_\infty$;
- $\{g_n\}$ is a Cauchy sequence in $B(E)$ as $\forall x \in E$ we have that:

  $$|g_n(x) - g_m(x)| = |f_n(x) - f_m(x)| \leqslant \|f_n - f_m\|_\infty \implies \|g_n - g_m\|_B \leqslant \|f_n - f_m\|_\infty.$$

$B(E)$ is complete (proposition 1.19):

$$\implies \exists\, g \in B(E) \text{ s.t. } \|g_n - g\|_B = \sup_{x \in E} |g_n(x) - g(x)| \xrightarrow[n\to\infty]{} 0.$$

Define $f(x) = \begin{cases} g(x) & x \in E \\ 0 & x \in E^c \end{cases}$

Then f is measurable as it's the limit a.e. of measurable functions.
$\forall x \notin E^c\ \lim_{n\to\infty} f_n(x) = \lim_{n\to\infty} g_n(x) = f(x)$.
It's easy to see that $f \in L^\infty$ as $f = \underbrace{(f - f_N)}_{\in L^\infty} + \underbrace{f_N}_{\in L^\infty} \in L^\infty$.

Let's prove that $\|f_n - f\|_\infty \xrightarrow[n\to\infty]{} 0$.

1. $x \in E \;\; |f_n(x) - f(x)| = |g_n(x) - g(x)| \leqslant \|g_n - g\|_B$;
2. $\mu(E^c) = 0 \implies |f_n(x) - f(x)| \leqslant \|g_n - g\|_B$ a.e.;

$$\implies \|f_n - f\|_\infty \leqslant \|g_n - g\|_B \to 0 \implies \|f_n - f\|_\infty \to 0.$$
$$\implies \forall \varepsilon > 0 \; \exists \, N : \|f_n - f\|_\infty < \varepsilon \; \forall n \geqslant N.$$

QED

Proposition 1.21. *L^p is a Banach space.*

Proof. Let $1 \leqslant p \lneqq \infty$.

We will use proposition 1.8. Let $\{f_n\} \in L^p$ s.t. $\sum \|f_n\|_p < \infty$. We want to show that $\sum f_n$ converges, that is there exists $F \in L^p$:

$$\left\| F - \sum_{m=1}^{k} f_m \right\|_p \xrightarrow{k \to \infty} 0.$$

We will prove this in two steps:

1. $\sum |f_m(x)|$ converges a.e.;
2. $\exists \; \lim_{k\to\infty} \sum_{m=1}^{k} f_m(x) =: F(x)$ a.e. and $\left\| F - \sum_{m=1}^{k} f_m \right\|_p \xrightarrow{k \to \infty} 0$.

Observe that $\forall x \in \Omega$ the sequence of partial sums

$$G_k(x) = \sum_{m=1}^{k} |f_m(x)|$$

is monotonic increasing $\implies \exists \; \lim_{k\to\infty} G_k(x)$. Let's prove that it's bounded a.e.. First we will prove that $G(x) = \lim_{k\to\infty} G_k(x) \implies G \in L^p$. It follows that $G(x)$ is bounded a.e.. Continuity of the function $t \mapsto t^p$ proves that

$$G(x)^p = \lim_{k\to\infty} G_k(x)^p \quad \forall x \in \Omega$$

We know that $\sum_{m=1}^{\infty} \|f_m\|_p = S < \infty$. Thus we can argue that:

$$\|G_k\|_p^p = \left\| \sum_{m=1}^{k} |f_m| \right\|_p^p \leqslant \left(\sum_{m=1}^{k} \|f_m\|_p \right)^p \leqslant S^p \implies \|G_k\|_p^p \leqslant S^p$$

$$\|G\|_p^p = \int_\Omega G^p d\mu = \int_\Omega \lim_{k\to\infty} G_k^p d\mu = {}^1 \lim_{k\to\infty} \int_\Omega G_k^p d\mu = \lim_{k\to\infty} \|G_k\|_p^p \leqslant S^p.$$

That is $\sum |f_m(x)| < \infty$ a.e. $\implies \sum f_m(x)$ converges a.e..

[1] Here we applied the Monotone convergence theorem.

Let's now show that $F(x) := \lim_{k\to\infty} \sum_{m=1}^{k} f_m(x)$ a.e.
$\implies F \in L^p$ and $\|F - \sum_{m=1}^{k} f_m\|_p \xrightarrow{k\to\infty} 0$.

$|F(x)| \leqslant \lim_{k\to\infty} \sum_{m=1}^{k} |f_m(x)| \leqslant \sum_{k=1}^{\infty} |f_m(x)| =: G(x)$ a.e.

$\implies |F(x)| \leqslant G(x)$ a.e. and $F \in L^p$.

$$\lim_{k\to\infty} \left\| \sum_{m=1}^{k} f_m - F \right\|_p = \lim_{k\to\infty} \left(\int_\Omega \left| \sum_{m=1}^{k} f_m - F \right|^p d\mu \right)^{1/p}. \tag{1.22}$$

Furthermore we have that:

$$\left| \sum_{n=1}^{k} f_n - F \right|^p \leqslant \left(\sum_{n=1}^{k} |f_n| + |F| \right)^p \leqslant (G+G)^p \text{ a.e. and } G^p \in L^1.$$

Then applying the dominated convergence theorem, we get:

$$(1.22) = \int_\Omega \lim_{k\to\infty} \left| \sum_{m=1}^{k} f_m - F \right|^p d\mu = 0.$$

We have proven that $\sum f_m$ converges in L^p.

QED

1.3 Hilbert spaces

Definition 1.23. Let $\mathscr{H}$ be a complex vector space. An inner product on $\mathscr{H}$ is a map $\langle\ ,\ \rangle:\ \mathscr{H}\times\mathscr{H}\longmapsto\mathbb{C}$ where $(\mathbf{x},\mathbf{y})\longmapsto\langle\mathbf{x},\mathbf{y}\rangle$ with the following properties:

1. $\langle a\cdot\mathbf{x}+b\cdot\mathbf{y},\mathbf{z}\rangle = a\cdot\langle\mathbf{x},\mathbf{z}\rangle + b\cdot\langle\mathbf{y},\mathbf{z}\rangle\,\forall a,b\in\mathbb{C},\ \mathbf{x},\mathbf{y},\mathbf{z}\in\mathscr{H}$;
2. $\langle\mathbf{x},\mathbf{y}\rangle = \overline{\langle\mathbf{y},\mathbf{x}\rangle}\ \forall\,\mathbf{x},\mathbf{y}\in\mathscr{H}$;
3. $\langle\mathbf{x},\mathbf{x}\rangle \gneqq 0\ \forall\,\mathbf{x}\in\mathscr{H}\backslash\{\mathbf{0}\}$.

Here we're going to state important results in the theory of Hilbert spaces, without proofs. All the notions are standard, and can be found in [Fol99].

Proposition 1.24. $\|\mathbf{v}\|_{\mathscr{H}} = \langle\mathbf{v},\mathbf{v}\rangle^{\frac{1}{2}}\ \forall\,\mathbf{v}\in\mathscr{H}$ *is a norm.*

Lemma 1.25. $|\langle\mathbf{u},\mathbf{v}\rangle| \leqslant \|\mathbf{u}\|\cdot\|\mathbf{v}\|\ \forall\,\mathbf{u},\mathbf{v}\in\mathscr{H}$.

Proposition 1.26. *Let* $(\mathscr{H},\|\cdot\|)$ *be a normed linear space:*
$\|\cdot\|$ *is induced by an inner product* $\Longleftrightarrow \|\mathbf{u}+\mathbf{v}\|^2+\|\mathbf{u}-\mathbf{v}\|^2 = 2\left(\|\mathbf{u}\|^2+\|\mathbf{v}\|^2\right)$.

Definition 1.27. $(\mathscr{H},\langle\cdot,\cdot\rangle)$ is a Hilbert space if $(\mathscr{H},\|\cdot\|_{\mathscr{H}})$ is a Banach space.

Definition 1.28. Let $\mathscr{H}$ be a Hilbert space. Consider $M\subset\mathscr{H}$: the define the distance of $\mathbf{v}\in\mathscr{H}$ from M as

$$d(\mathbf{v},M) = \inf_{\mathbf{w}\in M}\|\mathbf{v}-\mathbf{w}\|.$$

The next result gives an important characterization of closed, convex sets in Hilbert spaces, which we will use in the second chapter.

Theorem 1.29. *Consider a Hilbert space* $\mathscr{H}$ *and* $\mathbf{v}\in\mathscr{H}$. $M\subset\mathscr{H}$ *closed and convex* $\Longrightarrow\ \exists!\ \mathbf{u}\in M:\ \|\mathbf{u}-\mathbf{v}\|_{\mathscr{H}} = d\left(\mathbf{v},M\right)$

Proof. First note that $\mathbf{v}\in M \Longrightarrow d(\mathbf{v},M) = \inf_{\mathbf{w}\in M}\|\mathbf{v}-\mathbf{w}\|_{\mathscr{H}} \leqslant \|\mathbf{v}-\mathbf{v}\|_{\mathscr{H}} = 0$.
Also $d(0,M-\mathbf{v}) = \inf_{\mathbf{w}\in M}\|\mathbf{v}-\mathbf{w}\|_{\mathscr{H}} = d(\mathbf{v},M)$. WLOG: let $\mathbf{v}=0$. We want to prove there exists $\mathbf{u}\in\mathscr{H}$ s.t. $\|\mathbf{u}\| = \inf_{\mathbf{w}\in M}\|\mathbf{w}\|_{\mathscr{H}}$, and $\mathbf{u}$ is unique.
Suppose that:

$$\exists\ \mathbf{u_1},\mathbf{u_2}\in\mathscr{H}\text{s.t.}\|\mathbf{u_1}\|_{\mathscr{H}} = \|\mathbf{u_2}\|_{\mathscr{H}} = \inf_{\mathbf{w}\in M}\|\mathbf{w}\|_{\mathscr{H}} = m.$$

Then

$$\|\mathbf{u_1}-\mathbf{u_2}\|^2_{\mathscr{H}} = 2\cdot\left(\|\mathbf{u_1}\|^2_{\mathscr{H}}+\|\mathbf{u_2}\|^2_{\mathscr{H}}\right) - 4\cdot\left\|\frac{\mathbf{u_1}+\mathbf{u_2}}{2}\right\|^2_{\mathscr{H}} \leqslant 4\cdot m^2+4\cdot m^2 = 0.$$

Let $\inf_{\mathbf{w}\in M}\|\mathbf{w}\|_{\mathscr{H}} = m$. Then $\forall n\in\mathbb{N}$ there exists $\mathbf{w}_n\in M$ s.t.

$$m\leqslant\|\mathbf{w}_n\|_{\mathscr{H}}\leqslant m+\frac{1}{n}\ \Longrightarrow\ \lim_{n\to\infty}\|\mathbf{w}_n\|_{\mathscr{H}} = m.$$

$\mathscr{H}$ is complete, therefore all Cauchy sequence in $\mathscr{H}$ converge. We want to prove $\{\mathbf{w}_n\}_{n\in\mathbb{N}}$ is a Cauchy sequence.

$$\|\mathbf{w}_n - \mathbf{w}_k\|^2_{\mathscr{H}} = 2 \cdot \left(\|\mathbf{w}_n\|^2_{\mathscr{H}} + \|\mathbf{w}_k\|^2_{\mathscr{H}}\right) - 4 \cdot \left\| \frac{\mathbf{w}_n + \mathbf{w}_k}{2} \right\|^2_{\mathscr{H}}$$

$$\leqslant$$

$$2 \cdot \left(\left(m + \frac{1}{n}\right)^2 + \left(m + \frac{1}{k}\right)^2 \right) - 4 \cdot m^2 \xrightarrow[n,k\to\infty]{} 0.$$

M is also closed:

$$\exists\, \mathbf{u} \in M \text{ s.t. } \|\mathbf{w}_n - \mathbf{u}\|_{\mathscr{H}} \xrightarrow[n\to\infty]{} 0 \text{ and } \|\mathbf{u}\|_{\mathscr{H}} = \lim_{n\to\infty} \|\mathbf{w}_n\|_{\mathscr{H}} = m.$$

QED

Chapter 2

Uniform convexity and the Modulus of convexity

In this chapter, we present two new properties of sets in normed linear spaces: uniform convexity and strict convexity. Uniform convexity and strict convexity are stronger definitions of convexity, but they're not equivalent properties in infinite dimensional normed linear spaces. We'll explore how they are linked, particularly in finite dimensional normed linear spaces, and why strict convexity gives us key insights needed to prove Nordlander's Theorem.

2.1 Uniform convexity and strict convexity

First, we'll define the Modulus of convexity, through which we'll define uniform convexity.

Definition 2.1. Let (X,$\|\cdot\|$) be a normed linear space. The **Modulus** of convexity is a function that is defined for every $\varepsilon \in [0,2]$ as:

$$\delta(\varepsilon) = \inf\left\{1 - \left\|\frac{\mathbf{x}+\mathbf{y}}{2}\right\| \ \textit{where}\ \mathbf{x},\mathbf{y} \in X,\ \|\mathbf{x}\|,\|\mathbf{y}\| \leqslant 1,\ \|\mathbf{x}-\mathbf{y}\| \geqslant \varepsilon\right\}. \tag{2.2}$$

Definition 2.3. Let (X,$\|\cdot\|$) be a normed linear space. Then $\|\cdot\|$ is **uniformly** convex if for every $\varepsilon \in (0,2)$: $\delta(\varepsilon) > 0$.

Definition 2.4. Let (X,$\|\cdot\|$) be a normed linear space. Then $\|\cdot\|$ is **strictly** convex if $\forall\, \mathbf{x},\mathbf{y} \in X\ s.t.\ \|\mathbf{x}\|,\|\mathbf{y}\| \leqslant 1$: $\|\frac{\mathbf{x}+\mathbf{y}}{2}\| \lneqq 1$.

Both uniform convexity and strict convexity imply that the unit ball for the given norm has to always bend for all points on the boundary, as it's not possible for the ball to have flat edges. Uniform convexity is the requirement that the curve of boundary points bends by a minimum amount.

Example 2.5. $\|\cdot\|_{L^p}$ $1 \lneqq p \lneqq \infty$ are uniformly convex.

We will need the following inequalities:

Proposition 2.6. *Let f,g be measurable functions. Then:*

1. $\|\frac{f+g}{2}\|_p^p + \|\frac{f-g}{2}\|_p^p \leqslant \frac{1}{2}\cdot\|f\|_p^p + \frac{1}{2}\cdot\|g\|_p^p$ *for* $p \geqslant 2$*;*
2. $\|\frac{f+g}{2}\|_p^q + \|\frac{f-g}{2}\|_p^q \leqslant \left(\frac{1}{2}\cdot\|f\|_p^p + \frac{1}{2}\cdot\|g\|_p^p\right)^{\frac{q}{p}}$ *for* $1 \lneqq p \lneqq 2,\ \frac{1}{p}+\frac{1}{q}=1$*;*

A proof for such inequalities can be found in [Cla36].

Consider $\|f\|_p, \|g\|_p \leqslant 1,\ \|f-g\|_p \geqslant \varepsilon$.

Now we use Clarkson's inequality for $p \geqslant 2$:

$$\left\|\frac{f+g}{2}\right\|_p^p + \frac{\varepsilon^p}{2^p} \leqslant \frac{1}{2}\cdot 1^p + \frac{1}{2}\cdot 1^p. \tag{2.7}$$

We have that:

$$\left\|\frac{f+g}{2}\right\|_p \leqslant \left(1-\frac{\varepsilon^p}{2^p}\right)^{\frac{1}{p}} \implies 1-\left\|\frac{f+g}{2}\right\|_p \geqslant 1-\left(1-\frac{\varepsilon^p}{2^p}\right)^{\frac{1}{p}}. \tag{2.8}$$

Note that the right side does not depend on the choice of f,g, therefore

$$\delta_{L^p}(\varepsilon) = \inf\left\{1-\left\|\frac{f+g}{2}\right\|_p,\ \|f\|_p, \|g\|_p \leqslant 1,\ \|f-g\|_p \geqslant \varepsilon\right\} \tag{2.9}$$

$$\geqslant$$

$$1-\left(1-\frac{\varepsilon^p}{2^p}\right)^{\frac{1}{p}} \tag{2.10}$$

Note that $1-\left(1-\frac{\varepsilon^p}{2^p}\right)^{\frac{1}{p}} \gneqq 0\ \forall\varepsilon \in (0,2)$. Therefore $\delta_{L^p}(\varepsilon) \gneqq 0\ \forall\varepsilon \in (0,2)$, proving that L^p is uniformly convex for $p \geqslant 2$.

For $1 \lneqq p \lneqq 2$:

$$\left\|\frac{f+g}{2}\right\|_p^q + \frac{\varepsilon^q}{2^q} \leqslant \left(\frac{1}{2}\cdot 1^p + \frac{1}{2}\cdot 1^p\right)^{\frac{q}{p}}. \tag{2.11}$$

Note that: $q = \frac{p}{p-1} \geqslant p$ for all $1 \lneqq p \lneqq 2 \implies q \geqslant 2$.

$$\left\|\frac{f+g}{2}\right\|_p \leqslant \left(1-\left(\frac{\varepsilon}{2}\right)^q\right)^{\frac{1}{q}} \implies 1-\left\|\frac{f+g}{2}\right\|_p \geqslant 1-\left(1-\left(\frac{\varepsilon}{2}\right)^q\right)^{\frac{1}{q}}. \tag{2.12}$$

Then, following the proof above for $p \geqslant 2$, we have that $\delta_{L^p}(\varepsilon) \gneqq 0\ \forall\varepsilon \in (0,2)$, proving that L^p is uniformly convex for $1 \lneqq p \lneqq 2$.

Proposition 2.13. *Let (X,$\|\cdot\|$) be a normed linear space. If $\|\cdot\|$ is **uniformly** convex, then $\|\cdot\|$ is **strictly** convex.*

Proof. $\delta(\varepsilon) > 0 \implies 1 - \|\frac{\mathbf{y}+\mathbf{x}}{2}\| > \delta(\varepsilon) > 0 \ \forall \|\mathbf{x}\|, \|\mathbf{y}\| \leqslant 1 \ , \forall \varepsilon \in (0,2)$
$\implies 1 - \|\frac{\mathbf{y}+\mathbf{x}}{2}\| \gneqq 0.$ QED

Example 2.14. Both l^1 and l^∞ are not strictly convex.

1. Choose $\varepsilon \geqslant \frac{1}{2}$, and $\mathbf{x} = \{(1/4, 3/4, 0, 0, ...)\}, \mathbf{y} = \{(1/2, 1/2, 0, 0, ...)\} \in l^1$. Then $\|\mathbf{x} - \mathbf{y}\|_{l^1} \geqslant \varepsilon \geqslant \frac{1}{2}$. The segment between $\mathbf{x}$ and $\mathbf{y}$ is entirely contained in $\{\mathbf{x} \in l^1 : \ \|\mathbf{x}\|_{l^1} = 1\}$. Then $\delta(\varepsilon) = 0 \implies l^1$ is not uniformly convex. Also note that the midpoint $\frac{\mathbf{x}+\mathbf{y}}{2}$ has norm 1: l^1 is not strictly convex $\underset{2.13}{\implies} l^1$ is not uniformly convex.

2. Choose $\varepsilon \geqslant \frac{1}{2}$, and $x = \{(1, -1/4, 0, 0, ...)\}, \mathbf{y} = \{(1, +1/4, 0, 0, ...)\} \in l^\infty$. Then $\|\mathbf{x} - \mathbf{y}\|_{l^\infty} \geqslant \varepsilon \geqslant \frac{1}{2}$. The segment between $\mathbf{x}$ and $\mathbf{y}$ is entirely contained in $\{\mathbf{x} \in l^\infty : \ \|\mathbf{x}\|_{l^\infty} = 1\}$. Then $\delta(\varepsilon) = 0 \implies l^\infty$ is not uniformly convex Also note that the midpoint $\frac{\mathbf{x}+\mathbf{y}}{2}$ has norm 1: l^∞ is not strictly convex $\underset{2.13}{\implies} l^\infty$ is not uniformly convex.

The opposite implication is not true, as there exists counterexamples in infinite dimensional normed linear spaces.

Example 2.15. Consider $\mathbf{x} = \{(x_n)\}_{n\in\mathbb{N}}$ and the norm $\|\cdot\|$ defined as follows:

$$\|\mathbf{x}\| = \|\mathbf{x}\|_{l^1} + \|\mathbf{x}\|_{l^2} = \sum_{n\in\mathbb{N}} |x_n| + \sqrt{\sum_{n\in\mathbb{N}} |x_n|^2}.$$

$\|\cdot\|$ is strictly convex but not uniformly convex.

Proof. $\|\cdot\|$ is strictly convex due to $\|\cdot\|_{l^2}$ being strictly convex. To prove $\|\cdot\|$ is not uniformly convex, consider the standard basis $\{e_i\}$ in the infinite-dimensional sequence space. We define a scaling factor a_n and two sequences $\{\mathbf{x}_n\}_{n\in\mathbb{N}}$, $\{\mathbf{y}_n\}_{n\in\mathbb{N}}$ such that:

$$a_n = \frac{1}{n + \sqrt{n}},$$

$$\mathbf{x}_n = a_n \sum_{i=1}^{n} e_i, \quad \mathbf{y}_n = a_n \sum_{i=n+1}^{2n} e_i.$$

- $\|\mathbf{x}_n\|_1 = \sum_{i=1}^{n} |a_n| = n \cdot a_n = \frac{n}{n+\sqrt{n}}$.
- $\|\mathbf{x}_n\|_2 = \sqrt{\sum_{i=1}^{n} a_n^2} = \sqrt{n} \cdot a_n = \frac{\sqrt{n}}{n+\sqrt{n}}$.

Summing these gives the total norm:

$$\|\mathbf{x}_n\| = \|\mathbf{x}_n\|_1 + \|\mathbf{x}_n\|_2 = \frac{n+\sqrt{n}}{n+\sqrt{n}} = 1.$$

For any given $\varepsilon \in (0,2)$, there exist a sufficiently large $n_0 \in \mathbb{N}$ s.t. $n \geqslant n_0$

$$\|\mathbf{x}_n - \mathbf{y}_n\|_1 + \|\mathbf{x}_n - \mathbf{y}_n\|_2 = \frac{2n}{n+\sqrt{n}} + \frac{\sqrt{2n}}{n+\sqrt{n}} \geqslant \varepsilon$$

as $\frac{2n}{n+\sqrt{n}} + \frac{\sqrt{2n}}{n+\sqrt{n}} \xrightarrow[n\to\infty]{} 2$.

Thus, $\|\mathbf{x}_n - \mathbf{y}_n\| \geqslant \varepsilon \ \forall n \geqslant n_0$. The total midpoint norm is:

$$\left\|\frac{\mathbf{x}_n + \mathbf{y}_n}{2}\right\| = \frac{n+\frac{\sqrt{2n}}{2}}{n+\sqrt{n}}, \lim_{n\to\infty}\left\|\frac{\mathbf{x}_n + \mathbf{y}_n}{2}\right\| = \lim_{n\to\infty}\frac{1+\frac{\sqrt{2}}{2\sqrt{n}}}{1+\frac{1}{\sqrt{n}}} = 1.$$

$\implies \delta(\varepsilon) = 0$, completing the proof. QED

However if we restrict ourself to finite dimensional spaces, we find that the two properties are equivalent. To prove this, we'll have to give a characterization of compactness for finite dimensional normed linear spaces. We can prove the following:

Proposition 2.16. *Let X be a vector space on $\mathbb{R}$ where $dim(X) = n < \infty$.*
Let $\|\cdot\|$ be a norm on X. Then all closed and bounded sets in $(X, \|\cdot\|)$ are compact.

We're going to prove the proposition in steps. We want to find a norm $\|\cdot\|_*$ with which to construct an isometry between $(X, \|\cdot\|_*)$ and $(\mathbb{R}^n, \|\cdot\|_2)$, where $\|\cdot\|_2$ is the euclidean norm.

remark 2.17. Let X be a vector space on $\mathbb{R}$ where $dim(X) = n < \infty$.
There exist an isomorphism $f : X \longrightarrow \mathbb{R}^n$ s.t. the pull-back norm $\|\cdot\|_f$ defines an isometry between $\left(X, \|\cdot\|_f\right)$ and $(\mathbb{R}^n, \|\cdot\|_2)$.

Proof. Let $f : (X, \|\cdot\|) \longrightarrow (\mathbb{R}^n, \|\cdot\|_2)$ be an isomorphism s.t. $x_j = f^{-1}(e_j)$ $j = 1, ..., n$ where $e_1, ..., e_n$ is the standard orthonormal base in $\mathbb{R}^n$.
Let $\|\cdot\|_f$ be the pull-back norm on X via $f \implies \|x\|_f = \|f(x)\|_2 \ \forall x \in X$. We have that $f : \left(X, \|\cdot\|_f\right) \longrightarrow (\mathbb{R}^n, \|\cdot\|_2)$ is a linear isometry. QED

To prove proposition 2.16, we'll need to first find one norm for which this property holds.

Lemma 2.18. *Let X be a vector space on $\mathbb{R}$ where $dim(X) = n < \infty$.*
Let $f : X \longrightarrow \mathbb{R}^n$ be a isomorphism s.t. $f : \left(X, \|\cdot\|_f\right) \longrightarrow (\mathbb{R}^n, \|\cdot\|_2)$ is an isometry.
Let $\|\cdot\|_f$ be the pull-back norm. Then all closed and bounded sets in $\left(X, \|\cdot\|_f\right)$ are compact.

Proof. We have that $f : \left(X, \|\cdot\|_f\right) \longrightarrow (\mathbb{R}^n, \|\cdot\|_2)$ is a linear surjective isometry $\implies f$ is a homeomorphism. Homeomorphism preserve compactness, and the image of a closed and bounded set via f is closed and bounded.
Therefore if a set C is closed and bounded in $\left(X, \|\cdot\|_f\right)$, then $f(C)$ is closed, bounded in $(\mathbb{R}^n, \|\cdot\|_2) \implies$ compact in $(\mathbb{R}^n, \|\cdot\|_2)$.
But $f^{-1}(f(C)) = C$ is the image of a compact set via a continuous function f^{-1}.
Therefore C is compact in $\left(X, \|\cdot\|_f\right)$. QED

If we can prove that any give norm $\|\cdot\|$ on X is equivalent to $\|\cdot\|_f$, we can prove that any closed and bounded set in $(X, \|\cdot\|)$ is compact, as it's compact in $\left(X, \|\cdot\|_f\right)$.

Lemma 2.19. *Let X be a vector space on $\mathbb{R}$ where $dim(X) = n < \infty$.*
Let $\|\cdot\|_1, \|\cdot\|_2$ be equivalent norms on X. Then a set C is compact in $(X, \|\cdot\|_1) \iff$ it's compact $(X, \|\cdot\|_2)$.

Proof. $\|\cdot\|_1$ and $\|\cdot\|_2$ generate the same topology, so every open covering of C has a finite covering of open balls in $(X, \|\cdot\|_1)$ and $(X, \|\cdot\|_2)$. QED

Let's prove that all norms are equivalent in a finite dimensional normed linear space.

Theorem 2.20. *Let X be a vector space space on $\mathbb{R}$ such that $dim(X) = n < \infty$. Given $\|\cdot\|_1, \|\cdot\|_2$ norms on X: there exist a constant $c \geqslant 1$ s.t. $\forall\, \mathbf{x} \in X$ we have:*

$$c^{-1} \cdot \|\cdot\|_2 \leqslant \|\cdot\|_1 \leqslant c \cdot \|\cdot\|_2.$$

Proof. Consider the function $f : X \longrightarrow \mathbb{R}^n$ built in the proof of 2.17. We want to prove that $\|\cdot\|$ is equivalent to $\|\cdot\|_f$.

$$\|\mathbf{x}\| = \|\sum \lambda_i \cdot \mathbf{x}_i\| \leqslant \sum |\lambda_i| \cdot \|\mathbf{x}_i\| \leqslant {}^{1}\sqrt{\left(\sum \|\mathbf{x}_i\|^2\right)\left(\sum |\lambda_i|^2\right)} = c_1 \cdot \|\mathbf{x}\|_f \qquad (2.21)$$

where $c_1 = \sqrt{(\sum \|\mathbf{x}_i\|^2)}$.
We have also proved that $\|\cdot\| : \left(X, \|\cdot\|_f\right) \longrightarrow \mathbb{R}$ is continuous.

S^{n-1} is closed and bounded in $\left(X, \|\cdot\|_f\right) \underset{2.18}{\implies} S^{n-1}$ is compact in $\left(X, \|\cdot\|_f\right)$.
Also $\|\cdot\| : \left(X, \|\cdot\|_f\right) \longrightarrow \mathbb{R}$ is continuous and positive on S^{n-1}: there exist $c_2 > 0$ s.t. $\|\mathbf{x}\|_f = 1 \implies \|\mathbf{x}\| \geqslant c_2 > 0$.
So $\forall \mathbf{x} \in X \backslash \{0\} : \|\frac{x}{\|\mathbf{x}\|_f}\| \geqslant c_2 \iff \|\mathbf{x}\| \geqslant c_2 \cdot \|\mathbf{x}\|_f$ QED

Given that $\|\cdot\|_f$ is equivalent to $\|\cdot\|$, we have that a closed and bounded sets C in $(X, \|\cdot\|)$ is also closed and bounded in $\left(X, \|\cdot\|_f\right)$, therefore C is compact in $\left(X, \|\cdot\|_f\right) \underset{2.19}{\implies} C$ is compact in $(X, \|\cdot\|)$.
This completes the proof of proposition 2.16.

We can finally prove that strict convexity implies uniform convexity in finite dimensional normed linear spaces.

[1] Use the Cauchy-Schwarz inequality: $\langle \mathbf{v}, \mathbf{w} \rangle \leqslant \|\mathbf{v}\|_2 \cdot \|\mathbf{w}\|_2$ where $\mathbf{v}_i = |\lambda_i|$, $\mathbf{w}_i = \|x_i\|$.

Proposition 2.22. *Let (X,$\|\cdot\|$) be a finite dimensional normed linear space. $\|\cdot\|$ is* ***uniformly*** *convex if and only if $\|\cdot\|$ is* ***strictly*** *convex.*

Proof. We've already proven that uniform convexity implies strict convexity in proposition 2.22. Let's prove the opposite implication.
Consider $g : Y_\varepsilon \longrightarrow \mathbb{R}$ such that $g(\mathbf{x},\mathbf{y}) = 1 - \|\frac{\mathbf{x}+\mathbf{y}}{2}\|$.
If we can prove that $Y_\varepsilon = \{(\mathbf{x},\mathbf{y}) \in X \times X,\ \|\mathbf{x}\|, \|\mathbf{y}\| \leqslant 1,\ \|\mathbf{x}-\mathbf{y}\| \geqslant \varepsilon\}$ is compact, then the image of Y_ε via the continuous function g is compact in $\mathbb{R}$
$\implies a = \min\left\{1 - \|\frac{\mathbf{x}+\mathbf{y}}{2}\| \textit{ where } \mathbf{x},\mathbf{y} \in X,\ \|\mathbf{x}\|, \|\mathbf{y}\| \leqslant 1,\ \|\mathbf{x}-\mathbf{y}\| \geqslant \varepsilon\right\} \lneqq \infty$,
and $a \gneqq 0$ due to strict convexity.
Y_ε is compact $\underset{2.16}{\iff}$ it's closed and bounded in $(X \times X, \|\cdot\|_{X\times X})$. We'll define onto $X \times X$ the norm with the following property: $\|(\mathbf{x_1},\mathbf{x_2})\|_{X\times X} = \|\mathbf{x_1}\|_X + \|\mathbf{x_2}\|_X$.
Let's prove that Y_ε is closed.
Let $\{\mathbf{x}_n\}_{n\in\mathbb{N}}, \{\mathbf{y}_n\}_{n\in\mathbb{N}}$ be convergent sequences in Y_ε.
$\|\mathbf{y}_n - \mathbf{y}\| \xrightarrow[n\to\infty]{} 0, \|\mathbf{x}_n - \mathbf{x}\| \xrightarrow[n\to\infty]{} 0 \implies \|\mathbf{y}\| \leqslant 1, \|\mathbf{x}\| \leqslant 1$ due to the unit ball being closed in X. We also have that $\|\mathbf{y}_n - \mathbf{x}_n\| \geqslant \varepsilon$.
$\varepsilon \leqslant \|\mathbf{y}_n - \mathbf{x}_n\| = \|\mathbf{y}_n - \mathbf{y} + \mathbf{y} - \mathbf{x} + \mathbf{x} - \mathbf{x}_n\| \leqslant \|\mathbf{y}_n - \mathbf{y}\| + \|\mathbf{y} - \mathbf{x}\| + \|\mathbf{x} - \mathbf{x}_n\|$
$\to 0 + \|\mathbf{y} - \mathbf{x}\| + 0$.
$\varepsilon \leqslant \|\mathbf{y} - \mathbf{x}\|$. This proves that $(\mathbf{x},\mathbf{y}) \in Y_\varepsilon$. Y_ε is bounded as $\|\mathbf{y}\| \leqslant 1, \|\mathbf{x}\| \leqslant 1 \implies \|(\mathbf{x},\mathbf{y})\|_{X\times X} = \|\mathbf{x}\|_X + \|\mathbf{y}\|_X \leqslant 1 + 1\ \ \forall\ (\mathbf{x},\mathbf{y}) \in Y_\varepsilon$.

QED

2.2 Strictly convex differentiable norms and Supporting Hyperplane

We will now discuss properties of smooth and strictly convex norms. In particular, we will study how vectors defining supporting hyperplanes for the unit ball behave for different points on the boundary. These properties are vital to prove that the gradient for a smooth, strictly convex norm is injective on the unit sphere.

Definition 2.23. Let S be a closed set in $\mathbb{R}^n$. Then a supporting hyperplane H in $s \in \partial S$ is a hyperplane with the following properties:

1. S is entirely contained in one of two closed half-spaces bounded by the hyperplane;
2. $H \cap S \neq \emptyset$.

A standard reference can be found in [GH96].

remark 2.24. Given a closed set $S \subset \mathbb{R}^n$, a vector N defines a supporting hyperplane in $\mathbf{x} \in \partial S$ if $\langle N, \mathbf{z} \rangle \leqslant \langle N, \mathbf{x} \rangle \ \forall \mathbf{z} \in S$, where $\langle\ ,\ \rangle$ is the standard inner product on $\mathbb{R}^n$.

Proposition 2.25. *Let the norm $\|\cdot\| : \mathbb{R}^n \longrightarrow [0,\infty)$ be a smooth function.*
Let $\mathbb{D} = \{\mathbf{x} \in \mathbb{R}^n : \|\mathbf{x}\| \leqslant 1\}$, $\mathbf{x}, \mathbf{y} \in S^{\,n-1} = \{\mathbf{x} \in \mathbb{R}^n : \|\mathbf{x}\| = 1\}$.
Then assuming:

1. $\mathbf{x} \neq 0 \implies \nabla\|\mathbf{x}\| \neq 0$,
2. $\|\cdot\|$ *is strictly convex;*

we have that $n_{\mathbf{x}}$ and $n_{\mathbf{y}}$, the unit vectors for $\nabla\|\cdot\|\Big|_{\mathbf{x}}$, $\nabla\|\cdot\|\Big|_{\mathbf{y}}$ respectively, define supporting lines in $\mathbf{x} \neq \mathbf{y}$. Furthermore $\mathbf{x} \neq \mathbf{y} \implies n_{\mathbf{x}} \neq n_{\mathbf{y}}$.

Let's prove the following lemma first:

Proposition 2.26. *Let $\|\cdot\|$ be a strictly convex norm.*
Let $\mathbb{D} = \{\mathbf{x} \in \mathbb{R}^n : \|\mathbf{x}\| \leqslant 1\}$, $S^{\,n-1} = \{\mathbf{x} \in \mathbb{R}^n : \|\mathbf{x}\| = 1\}$. Then there exist $N_{\mathbf{x}}$ definining a supporting line in $\mathbf{x}\ \forall\ \mathbf{x} \in S^{\,n-1}$.
Let $\mathbf{x}, \mathbf{y} \in S^{\,n-1}$. We have that $\mathbf{x} \neq \mathbf{y} \implies N_{\mathbf{x}} \neq N_{\mathbf{y}}$.

Proof. $\|\cdot\|$ strictly convex $\implies \mathbb{D}$ is convex. Also $\mathbb{D}$ is closed.
Let $\{\mathbf{y}_n\}_{n\in\mathbb{N}} \subset \mathbb{R}^n \backslash \mathbb{D}$ be a sequence that converges to a point $\mathbf{x} \in S^{\,n-1}$. Such a sequence exist because $\partial(\mathbb{R}^n \backslash \mathbb{D}) = \partial(\mathbb{D}) = S^{\,n-1}$. Let $\mathbf{x}_n = P(\mathbf{y}_n)$ be a projection of $\mathbf{y}_n$ onto $\mathbb{D}\ \forall n \in \mathbb{N}$. $\mathbf{x}_n$ is defined as the vector in $\mathbb{D}$ that satisfies

$$\min_{\mathbf{v}\in\mathbb{D}} \|\mathbf{y}_n - \mathbf{v}\| = \|\mathbf{x}_n - \mathbf{y}_n\| \ \forall n \in \mathbb{N}. \tag{2.27}$$

We also have that:

1. $\langle \mathbf{y}_n - \mathbf{x}_n, \mathbf{z} - \mathbf{x}_n \rangle \leqslant 0 \; \forall \mathbf{z} \in \mathbb{D}$.
2. $0 \leqslant \|\mathbf{x}_n - \mathbf{x}\| = \|P(\mathbf{y}_n) - P(\mathbf{x})\| \leqslant \|\mathbf{y}_n - \mathbf{x}\| \xrightarrow[n\to\infty]{} 0$.

Let $\mathbf{N}_{\mathbf{x},n} = \frac{\mathbf{y}_n - \mathbf{x}_n}{\|\mathbf{y}_n - \mathbf{x}_n\|}$, then $\langle \mathbf{N}_{\mathbf{x},n}, \mathbf{z} - \mathbf{x}_n \rangle \leqslant 0 \; \forall \mathbf{z} \in \mathbb{D}$ $\{\mathbf{N}_{\mathbf{x},n}\}_{n\in\mathbb{N}} \in S^{\,n-1}$.
$S^{\,n-1}$ is compact in $\mathbb{R}^n$, therefore there exist a convergent subsequence $\{\mathbf{N}_{\mathbf{x},n_k}\}_{k\in\mathbb{N}}$.

$$0 \leqslant |\, \langle \mathbf{N}_{\mathbf{x},n_k}, \mathbf{z} - \mathbf{x}_{n_k} \rangle - \langle \mathbf{N}, \mathbf{z} - \mathbf{x} \rangle \,|$$

$$\leqslant |\, \langle \mathbf{N}_{\mathbf{x},n_k}, \mathbf{z} - \mathbf{x}_{n_k} \rangle - \langle \mathbf{N}_{\mathbf{x},n_k}, \mathbf{z} - \mathbf{x} \rangle \,| + |\, \langle \mathbf{N}_{\mathbf{x},n_k}, \mathbf{z} - \mathbf{x} \rangle - \langle \mathbf{N}, \mathbf{z} - \mathbf{x} \rangle \,|$$

$$^{2} \leqslant \tilde{C}\Big[\|\mathbf{N}_{\mathbf{x},n_k}\| \cdot \|\mathbf{x}_{n_k} - \mathbf{x}\| + \|\mathbf{N}_{\mathbf{x},n_k} - \mathbf{N}\| \cdot \|\mathbf{z} - \mathbf{x}\| \Big] \xrightarrow[k\to\infty]{} 0.$$

Therefore $\langle \mathbf{N}_{\mathbf{x},n_k}, \mathbf{z} - \mathbf{x}_{n_k} \rangle \xrightarrow[k\to\infty]{} \langle \mathbf{N}, \mathbf{z} - \mathbf{x} \rangle$, and continuity implies

$$\langle \mathbf{N}, \mathbf{z} - \mathbf{x} \rangle \leqslant 0 \; \forall \mathbf{z} \in \mathbb{D}.$$

The vector $\mathbf{N}$ defines a supporting line in $\mathbf{x}$.

Let's assume that $\mathbf{N} = \mathbf{N}_{\mathbf{x}} = \mathbf{N}_{\mathbf{y}}$ defines a supporting line in $\mathbf{x}$ and $\mathbf{y}$, where $\mathbf{x} \neq \mathbf{y}$ and $\|\mathbf{N}\| = 1$.

$$\langle \mathbf{N}, \mathbf{y} - \mathbf{x} \rangle = \langle \mathbf{N}_{\mathbf{x}}, \mathbf{y} - \mathbf{x} \rangle = -\left\langle \mathbf{N}_{\mathbf{y}}, \mathbf{x} - \mathbf{y} \right\rangle = -\langle \mathbf{N}, \mathbf{y} - \mathbf{x} \rangle$$
$$\implies \langle \mathbf{N}, \mathbf{y} - \mathbf{x} \rangle = 0 \implies \langle \mathbf{N}, \theta\mathbf{y} + (1-\theta)\,\mathbf{x} \rangle = 0 \; \forall \theta \in [0,1]\,.$$

Therefore $\theta\mathbf{y} + (1-\theta)\,\mathbf{x} \in S^{\,n-1} \; \forall \theta \in [0,1]$.
For $\theta = \frac{1}{2}$: $\left\langle \mathbf{N}, \frac{\mathbf{y}+\mathbf{x}}{2} \right\rangle = 0 \implies \frac{\mathbf{y}+\mathbf{x}}{2} \in S^{\,n-1}$, but $\|\frac{\mathbf{y}+\mathbf{x}}{2}\| \lneqq 1$ as $\|\cdot\|$ is strictly convex. $\lightning$
QED

We need to prove that the $\nabla\|\mathbf{x}\|$ defines a supporting line in $\mathbf{x}$.

Let's consider the function $\mathbf{x} \longmapsto \|\mathbf{x}\|$:

$$\|\theta\mathbf{x} + (1-\theta)\,z\| \leqslant \theta\|\mathbf{x}\| + (1-\theta)\,\|\mathbf{z}\| \;\; \forall\, \mathbf{x}, \mathbf{z} \in \mathbb{D},\; \forall \theta \in [0,1]\,.$$

The function is convex on a convex domain $\mathbb{D}$.

This implies $\|\mathbf{z}\| - \|\mathbf{x}\| \geqslant \langle \nabla\|\mathbf{x}\|, \mathbf{z} - \mathbf{x} \rangle \;\; \forall\, \mathbf{y} \in \mathbb{D}, \mathbf{x} \in S^{\,n-1}$.

$$\implies 0 \geqslant \|\mathbf{z}\| - 1 \geqslant \langle \nabla\|\mathbf{x}\|, \mathbf{z} - \mathbf{x} \rangle \, \forall\, \mathbf{z} \in \mathbb{D}$$

$\nabla\|\mathbf{x}\|$ defines a supporting line in $\mathbf{x}$. Proposition 2.26 implies that $\nabla\|\mathbf{x}\| \neq \nabla\|\mathbf{y}\|$ for $\mathbf{x} \neq \mathbf{y}$.

[2] All norms in $\mathbb{R}^2$ are equivalent. There exist a constant $\tilde{C} \in \mathbb{R}_+$ s.t. $\sqrt{\langle\, x, x \,\rangle} = \|x\|_2 \leqslant \tilde{C}\|x\|$

Chapter 3

Nordlander's Theorem

In this chapter we provide a proof of Nordlander's Theorem. We show that for any normed linear space, the modulus of convexity is bounded by the modulus of convexity of a Hilbert space.

3.1 Modulus of convexity for a Hilbert space

To calculate the modulus of convexity for a Hilbert space we'll use the parallelogram law. Let's consider a Hilbert space $(H, \|\cdot\|_H)$ where $\|\cdot\|_H$ satisfies the following property:

$$\|\mathbf{x}+\mathbf{y}\|_H^2 + \|\mathbf{x}-\mathbf{y}\|_H^2 = 2\cdot\|\mathbf{y}\|_H^2 + 2\cdot\|\mathbf{x}\|_H^2 \;\; \forall \mathbf{x},\mathbf{y} \in H. \tag{3.1}$$

Let $\|\mathbf{x}\|_H, \|\mathbf{y}\|_H \leqslant 1$ s.t. s.t. $\|\mathbf{x}-\mathbf{y}\|_H \geqslant \varepsilon$.

$$\implies \|\mathbf{x}+\mathbf{y}\|_H^2 + \varepsilon^2 \leqslant \|\mathbf{x}+\mathbf{y}\|_H^2 + \|\mathbf{x}-\mathbf{y}\|_H^2 \leqslant 2\cdot 1 + 2\cdot 1.$$

$$\implies 1 - \|\frac{\mathbf{x}+\mathbf{y}}{2}\|_H \geqslant 1 - \sqrt{1-\frac{\varepsilon^2}{4}}. \tag{3.2}$$

Consider $\mathbf{x},\mathbf{y} \in S^1$ s.t. $\|\mathbf{x}-\mathbf{y}\| = \varepsilon$.
Substituting these values in 3.1, we have:

$$\|\mathbf{x}+\mathbf{y}\|_H^2 + \varepsilon^2 = 2\cdot 1 + 2\cdot 1 \iff \|\frac{\mathbf{x}+\mathbf{y}}{2}\|_H^2 + \frac{\varepsilon^2}{4} = 1 \iff \|\frac{\mathbf{x}+\mathbf{y}}{2}\|_H = \sqrt{1-\frac{\varepsilon^2}{4}}.$$

$$\implies 1 - \|\frac{\mathbf{x}+\mathbf{y}}{2}\|_H = 1 - \sqrt{1-\frac{\varepsilon^2}{4}}. \tag{3.3}$$

We have that:

$$\delta_H(\varepsilon) = \inf\left\{1 - \|\frac{\mathbf{x}+\mathbf{y}}{2}\|_H,\ \|\mathbf{x}\|,\|\mathbf{y}\| \leqslant 1:\ \|\mathbf{x}-\mathbf{y}\|_H \geqslant \varepsilon\right\} = 1 - \sqrt{1-\frac{\varepsilon^2}{4}}. \tag{3.4}$$

3.2 Main Statement

Here we state Nordlander's theorem [Nor60].

Theorem 3.5. *Let $(X, \|\cdot\|)$ be a normed linear space over $\mathbb{R}$. Let $\delta_X(\varepsilon)$ be the modulus of convexity for $(X, \|\cdot\|)$ for $\varepsilon \in [0,2]$. Then*

$$\delta_X(\varepsilon) \leqslant \delta_H(\varepsilon) = 1 - \sqrt{1 - \frac{\varepsilon^2}{4}} \ \ \forall \varepsilon \in [0,2]$$

where H is any Hilbert space.

3.3 Proof for a 2D subspace

We're going to prove the inequality for any 2-Dimensional subspace within X. Observe that:

$$\inf\left\{1 - \|\frac{\mathbf{x}+\mathbf{y}}{2}\|,\ \mathbf{x},\mathbf{y} \in X \ s.t\ \|\mathbf{x}\|, \|\mathbf{y}\| \leqslant 1 :\ \|\mathbf{x}-\mathbf{y}\| \geqslant \varepsilon\right\}$$

$$=$$

$$\inf_{\substack{V \subseteq X \ subspace, \\ dim(V)=2}} \inf\left\{1 - \|\frac{\mathbf{x}+\mathbf{y}}{2}\|,\ \mathbf{x},\mathbf{y} \in V \ s.t\ \|\mathbf{x}\|, \|\mathbf{y}\| \leqslant 1 :\ \|\mathbf{x}-\mathbf{y}\| \geqslant \varepsilon\right\}.$$

If the inequality is true for any arbitrary 2D subspace of X, then it holds for X.

Let's consider a 2D subspace V. To prove the inequality, we'll first consider an equivalent definition of the modulus of convexity.

Lemma 3.6. *For $\mathbf{x},\mathbf{y} \in V$:*

$$\delta_V(\varepsilon) = \inf\left\{1 - \left\|\frac{\mathbf{x}+\mathbf{y}}{2}\right\| :\ \|\mathbf{x}\| = \|\mathbf{y}\| = 1,\ \|\mathbf{x}-\mathbf{y}\| = \varepsilon\right\}.$$

Proof. Let:

$$A = \left\{1 - \left\|\frac{\mathbf{x}+\mathbf{y}}{2}\right\| :\ \ \|\mathbf{x}\|, \|\mathbf{y}\| \leqslant 1,\ \|\mathbf{x}-\mathbf{y}\| \geqslant \varepsilon\right\},$$

$$B = \left\{1 - \left\|\frac{\mathbf{x}+\mathbf{y}}{2}\right\| :\ \|\mathbf{x}\| = \|\mathbf{y}\| = 1,\ \|\mathbf{x}-\mathbf{y}\| = \varepsilon\right\}.$$

$B \subseteq A \implies \inf(A) \leqslant \inf(B)$.
For the opposite implication, consider the function $g(\mathbf{x},\mathbf{y}) = \|\frac{\mathbf{x}+\mathbf{y}}{2}\| \ \forall \|\mathbf{x}\|, \|\mathbf{y}\| \leqslant 1$. This function is convex and is defined on a compact (see proposition 2.16) and convex set

$$\{(\mathbf{x},\mathbf{y}) \in V \times V : \|\mathbf{x}\|, \|\mathbf{y}\| \leqslant 1\}.$$

Consider $\|\mathbf{z}\| \lneqq 1 \implies \mathbf{z} = \theta \cdot \mathbf{z}_1 + (1-\theta) \cdot \mathbf{z}_2$ for some $\theta \in (0,1)$, $\|\mathbf{z}_1\| = \|\mathbf{z}_2\| = 1$.

Let $M = \max(g(\mathbf{z}_1), g(\mathbf{z}_2))$, then $g(\mathbf{z}) \leqslant \theta \cdot g(\mathbf{z}_1) + (1-\theta) \cdot g(\mathbf{z}_2) \leqslant M$

$$\implies \max_{\|\mathbf{x}\|,\|\mathbf{y}\| \leqslant 1} g(\mathbf{x},\mathbf{y}) = \max_{\|\mathbf{x}\|=\|\mathbf{y}\|=1} g(\mathbf{x},\mathbf{y}).$$

We've proven that $\inf(A) = \inf\left\{1 - \left\|\frac{\mathbf{x}+\mathbf{y}}{2}\right\| : \ \|\mathbf{x}\| = \|\mathbf{y}\| = 1, \ \|\mathbf{x}-\mathbf{y}\| \geqslant \varepsilon\right\}$.

We want to prove that there exist $\|\mathbf{x}'\|, \|\mathbf{y}'\| \leqslant 1$ s.t.

$$\left\|\frac{\mathbf{x}'+\mathbf{y}'}{2}\right\| \geqslant \left\|\frac{\mathbf{x}+\mathbf{y}}{2}\right\|, \ \varepsilon = \|\mathbf{x}'-\mathbf{y}'\| \lneqq \|\mathbf{x}-\mathbf{y}\| = A, \|\mathbf{x}\|, \|\mathbf{y}\| \leqslant 1.$$

Define $\mathbf{m} = \frac{\mathbf{x}+\mathbf{y}}{2}$, $\mathbf{v} = \frac{\mathbf{y}-\mathbf{x}}{2}$, then $\mathbf{x} = \mathbf{m} - \frac{\mathbf{v}}{2}$, $\mathbf{y} = \mathbf{m} + \frac{\mathbf{v}}{2}$.
Choose $\mathbf{x}' = \mathbf{m} - \frac{\varepsilon}{A} \cdot \frac{\mathbf{v}}{2}$, $\mathbf{y}' = \mathbf{y} + \frac{\varepsilon}{A} \cdot \frac{\mathbf{v}}{2}$.

It's clear that $\mathbf{x}'$, $\mathbf{y}'$ are contained in the segment joining $\mathbf{x}$, $\mathbf{y}$ and
$\|\mathbf{y}' - \mathbf{x}'\| = \|\frac{\varepsilon}{A} \cdot \mathbf{v}\| = \varepsilon$.

The midpoint $\mathbf{x}'$ and $\mathbf{y}'$ is $\mathbf{m} \implies \left\|\frac{\mathbf{x}'+\mathbf{y}'}{2}\right\| = \left\|\frac{\mathbf{x}+\mathbf{y}}{2}\right\|$. This holds for $\|\mathbf{x}\| = \|\mathbf{y}\| = \|\mathbf{x}'\| = \|\mathbf{y}'\| = 1$. We have proven that $\inf(A) \not< \inf(B)$. $\implies \inf(A) = \inf(B)$. QED

3.4 Proving the inequality

Let $T : V \longrightarrow \mathbb{R}^2$ be an isomorphism s.t. $x_j = T^{-1}(e_j)$ for $j = 1,2$ where e_1, e_2 is the standard orthonormal base in $\mathbb{R}^2$. T is a diffeomorphism, and $\|x\|_{T^{-1}} = \|T^{-1}(x)\|$ is the pull-back norm defined on $\mathbb{R}^2$.
$T : (V, \|\cdot\|) \longrightarrow \left(\mathbb{R}^2, \|\cdot\|_{T^{-1}}\right)$ is an isometry.

Let $\mathbf{Y}$ be the area enclosed within $S^1 = \Gamma = \{\mathbf{y} \in \mathbb{R}^2 : \|\mathbf{y}\|_{T^{-1}} = 1\}$, and let $\mathbf{Y}_\varepsilon$ be the area enclosed within $\Gamma_\varepsilon = \left\{\frac{\mathbf{x}+\mathbf{y}}{2} : \ \mathbf{x},\mathbf{y} \in S^1, \ \|\mathbf{x}-\mathbf{y}\|_{T^{-1}} = \varepsilon\right\}$.

Lemma 3.7. $\mathbf{Y}_\varepsilon = \left(1 - \frac{\varepsilon^2}{4}\right)\mathbf{Y}$.

The area within Γ is calculate as

$$\int_{\{\|\mathbf{x}\|_{T^{-1}} \leqslant 1 : \ \mathbf{x} \in \mathbb{R}^2\}} dx\, dy$$

We want to express the integral in polar coordinates:

consider $\begin{cases} x = r \cdot \cos(\phi) \\ y = r \cdot \sin(\phi) \end{cases}$ for $\phi \in [0, 2\pi]$, $r \in [0, \infty)$.

For any $\mathbf{v}, \mathbf{w} \in V$ s.t. $\|\mathbf{v} - \mathbf{w}\| = \varepsilon$, there exist $\mathbf{x} = T(\mathbf{v}), \mathbf{y} = T(\mathbf{w})$ s.t. $\|T(\mathbf{v} - \mathbf{w})\|_T^{-1} = \|\mathbf{v} - \mathbf{w}\| = \varepsilon$.
For all $\mathbf{z}$ s.t. $\|\mathbf{z}\|_{T^{-1}} = 1$, let $\mathbf{z}_x = \tilde{r} \cdot \cos(\phi)$, $\mathbf{z}_y = \tilde{r} \cdot \sin(\phi)$, with $\tilde{r} := \|\mathbf{z}\|_2$, where $\|\cdot\|_2$ is the euclidean norm. $\tilde{r}$ is not fixed, and it varies according to ϕ.
Let $\mathbf{z}(\phi) = (\mathbf{z}_x, \mathbf{z}_y)$ as defined above, with $\|\mathbf{z}(\phi)\|_2 = \|\mathbf{z}\|_2 = \tilde{r}$.

Let $\mathbf{A}(\phi), \mathbf{B}(\phi)$ s.t. $\|\mathbf{A}(\phi)\|_{T^{-1}} = \|\mathbf{B}(\phi)\|_{T^{-1}} = 1$, $\|\mathbf{A} - \mathbf{B}\|_{T^{-1}} = \varepsilon$.

$$\mathbf{Y} = \int_{\{\|\mathbf{x}\|_{T^{-1}} \leqslant 1:\ \mathbf{x} \in \mathbb{R}^2\}} dx\, dy = \int_0^{2\pi} \int_0^{\|\mathbf{A}(\phi)\|_2} \rho\, d\rho\, d\phi. \tag{3.8}$$

$$\mathbf{Y}_\varepsilon = \int_0^{2\pi} \int_0^{\|\frac{\mathbf{A}(\phi)+\mathbf{B}(\phi)}{2}\|_2} \rho\, d\rho\, d\phi = \int_0^{2\pi} \frac{\left\|\frac{\mathbf{A}(\phi)+\mathbf{B}(\phi)}{2}\right\|_2^2}{2} d\phi \tag{3.9}$$

$$\mathbf{Y} - \mathbf{Y}_\varepsilon = \int_0^{2\pi} \left[\frac{\|\mathbf{A}(\phi)\|_2^2}{2} - \frac{\left\|\frac{\mathbf{A}(\phi)+\mathbf{B}(\phi)}{2}\right\|_2^2}{2} \right] d\phi$$

$\left(\mathbb{R}^2, \langle \cdot, \cdot \rangle\right)$ is a Hilbert space: $\|\mathbf{A}(\phi)\|_2^2 = \langle \mathbf{A}(\phi), \mathbf{A}(\phi) \rangle$

$$\frac{1}{2} \cdot \|\mathbf{A}(\phi)\|_2^2 - \left\| \frac{\mathbf{A}(\phi) + \mathbf{B}(\phi)}{2} \right\|_2^2 = -\frac{1}{2} \cdot \|\mathbf{B}(\phi)\|_2^2 + \left\| \frac{\mathbf{A}(\phi) - \mathbf{B}(\phi)}{2} \right\|_2^2$$

$$\mathbf{Y} - 2 \cdot \mathbf{Y}_\varepsilon = \int_0^{2\pi} \frac{\|\mathbf{A}(\phi)\|_2^2}{2} - \left\| \frac{\mathbf{A}(\phi) + \mathbf{B}(\phi)}{2} \right\|_2^2 d\phi = \int_0^{2\pi} -\frac{\|\mathbf{B}(\phi)\|_2^2}{2} + \left\| \frac{\mathbf{A}(\phi) - \mathbf{B}(\phi)}{2} \right\|_2^2 d\phi.$$

$$\mathbf{X} := \int_0^{2\pi} \frac{\|\mathbf{A}(\phi) - \mathbf{B}(\phi)\|_2^2}{2} d\phi = \int_0^{2\pi} \int_0^{\|\mathbf{A}(\phi) - \mathbf{B}(\phi)\|_2} \rho\, d\rho\, d\phi$$

$$\mathbf{Y} = \int_0^{2\pi} \frac{\|\mathbf{A}(\phi)\|_2^2}{2} d\phi = \int_0^{2\pi} \frac{\|\mathbf{B}(\phi)\|_2^2}{2} d\phi$$

$$\mathbf{Y} - \mathbf{Y}_\varepsilon = \mathbf{Y} - 2 \cdot \mathbf{Y}_\varepsilon + \mathbf{Y}_\varepsilon = -\mathbf{Y} + \frac{1}{2} \cdot \mathbf{X} + \mathbf{Y}_\varepsilon$$

$$\mathbf{Y} - \mathbf{Y}_\varepsilon = \frac{1}{4} \cdot \mathbf{X} = \frac{1}{4} \left[\int_0^{2\pi} \int_0^{\|\mathbf{A}(\phi) - \mathbf{B}(\phi)\|_2} \rho\, d\rho\, d\phi \right]. \tag{3.10}$$

To calculate the integral on the right hand side of (3.10), we will assume that $\|\cdot\|\big|_V$ is strictly convex and differentiable.

Let $\alpha(\phi)$ be the angle between the ray $y \equiv \mathbf{A}_y$ starting in $\mathbf{A}(\phi)$ parallel to the x-axis and $\mathbf{B}(\phi) - \mathbf{A}(\phi)$.
Consider $\mathbf{v}$, the vector with starting point $\mathbf{A}(\phi)$ and endpoint $\mathbf{B}(\phi)$.
$\|\mathbf{v}(\alpha(\phi))\|_2 = \|\mathbf{A}(\phi) - \mathbf{B}(\phi)\|_2 \ \forall \phi \in [0, 2\pi]$. The integral $\mathbf{X}$ is the area within the curve $\Gamma_{\mathbf{v}}$ drawn by the endpoint of $\mathbf{v}(\alpha(\phi))$. $\Gamma_{\mathbf{v}}$ is similar to Γ, where the radius in polar coordinates is scaled by a factor c.

If we can prove the following:

1. $\alpha(\phi)$ is injective and differentiable;
2. $c = \varepsilon$;

then

$$X = \int_0^{2\pi} \int_0^{\|\mathbf{A}(\phi)-\mathbf{B}(\phi)\|_2} \rho \, d\rho \, d\phi = \int_0^{2\pi} \int_0^{\varepsilon \cdot \|\mathbf{A}(\phi)\|_2} \rho \cdot d\rho \, d\phi = \varepsilon^2 \cdot Y \quad (3.11)$$

Note that:

$$c \equiv \frac{\|\mathbf{B}(\phi) - \mathbf{A}(\phi)\|_2}{\|\mathbf{A}(\phi)\|_2} \equiv \frac{\|\mathbf{B}(\phi) - \mathbf{A}(\phi)\|_{T^{-1}}}{\|\mathbf{A}(\phi)\|_{T^{-1}}} \equiv \varepsilon \ \forall \phi \in [0, 2\pi].$$

Therefore

$$\mathbf{Y} - \mathbf{Y}_\varepsilon = \frac{1}{4} \cdot \varepsilon^2 \cdot \mathbf{Y} \implies \mathbf{Y}_\varepsilon = \left(1 - \frac{\varepsilon^2}{4}\right) \mathbf{Y}. \quad (3.12)$$

3.4.1 $\alpha(\phi)$ is differentiable

To prove that $\alpha(\phi)$ is differentiable, we'll need the following observation.

If we can prove that $\mathbf{A}(\phi)$ is differentiable, then $\alpha(\phi)$ is differentiable. We want to apply the implicit function theorem.
Lets consider the function $\mathbf{F} : [0, 2\pi] \times \mathbb{R}^2 \to \mathbb{R}^2$ such that:

$$\mathbf{F}(\phi, \mathbf{y}) = [\|\mathbf{y}\|_{T^{-1}} - 1, \|\mathbf{y} - \mathbf{B}(\phi)\|_{T^{-1}} - \varepsilon] \tag{3.13}$$

where $\mathbf{y} \in \mathbb{R}^2$ and $\mathbf{B}(\phi)$ is a point on the unit ball ($\|\mathbf{B}(\phi)\|_{T^{-1}} = 1$).

$\forall\, \tilde{\mathbf{y}} \in \mathbb{R}^2$ s.t. $\|\tilde{\mathbf{y}}\|_{T^{-1}} = 1$ and $\|\tilde{\mathbf{y}} - \mathbf{B}(\phi)\|_{T^{-1}} = \varepsilon$:

$$\mathbf{F}(\phi, \tilde{\mathbf{y}}) \equiv \mathbf{0}\ \forall \phi \in [0, 2\pi]\,. \tag{3.14}$$

$$\mathbb{J}_{(\phi,\mathbf{y})}(\mathbf{F}) = \left[\begin{array}{c|cc} \dfrac{\partial F_1}{\partial \phi} & \dfrac{\partial F_1}{\partial x} & \dfrac{\partial F_1}{\partial y} \\ \dfrac{\partial F_2}{\partial \phi} & \dfrac{\partial F_2}{\partial x} & \dfrac{\partial F_2}{\partial y} \end{array}\right] = \left[\mathbb{J}_{\phi}(\mathbf{F}) \,\middle|\, \mathbb{J}_{(x,y)}(\mathbf{F})\right]$$

The implicit function theorem implies the existence of a function $\mathbf{A}(\phi)$ defined in a neighborhood $U_{\bar{\phi}}$ of $\bar{\phi}$ such that $\mathbf{F}(\phi, A(\phi)) \equiv \mathbf{0}\ \forall \phi \in U_{\bar{\phi}}$.
Let $\mathbb{J}_{\mathbf{x}}(\mathbf{F})$ be the Jacobian of the function $\mathbf{F}$ with respect to the variables $\mathbf{x}$.

$$\mathbf{0} \equiv \mathbb{J}_{(\phi,\mathbf{y})}(\mathbf{F})(\phi, \mathbf{A}(\phi)) = \left[\mathbb{J}_{\phi}(\mathbf{F})(\phi, \mathbf{A}(\phi)) \,\middle|\, \mathbb{J}_{(x,y)}(\mathbf{F})(\phi, \mathbf{A}(\phi))\right] \cdot \left[\begin{array}{c} \mathbb{J}_{\phi}(\phi) \\ \hline \mathbb{J}_{\phi}(\mathbf{A}(\phi)) \end{array}\right]$$

$$= \left[\begin{array}{c|cc} \partial_{\phi}\|\mathbf{y}\|_{T^{-1}} & \partial_x\|\mathbf{y}\|_{T^{-1}} & \partial_y\|\mathbf{y}\|_{T^{-1}} \\ \partial_{\phi}\|\mathbf{y} - \mathbf{B}(\phi)\|_{T^{-1}} & \partial_x\|\mathbf{y} - \mathbf{B}(\phi)\|_{T^{-1}} & \partial_y\|\mathbf{y} - \mathbf{B}(\phi)\|_{T^{-1}} \end{array}\right] \cdot \left[\begin{array}{c} 1 \\ \hline \partial_{\phi}A_1(\phi) \\ \partial_{\phi}A_2(\phi) \end{array}\right]$$

$$= \mathbb{J}_{\phi}(\mathbf{F})(\phi, A(\phi)) + \mathbb{J}_{(x,y)}(\mathbf{F})(\phi, A(\phi)) \cdot \mathbb{J}_{\phi}(A)(\phi) \tag{3.15}$$

To prove that $\mathbf{A}(\phi)$ is differentiable, we must first prove that $\mathbb{J}_{(x,y)}(\mathbf{F})(\phi, \mathbf{A}(\phi))$ is invertible:

$$\mathbb{J}_{\phi}(\mathbf{A})(\phi) = -\mathbb{J}^{-1}_{(x,y)}(\mathbf{F})(\phi, \mathbf{A}(\phi)) \cdot \mathbb{J}_{\phi}(\mathbf{F})(\phi, \mathbf{A}(\phi))\,.$$

$$\mathbb{J}_{(x,y)}(\mathbf{F}) = \begin{bmatrix} \partial_x\|\mathbf{y}\|_{T^{-1}} & \partial_y\|\mathbf{y}\|_{T^{-1}} \\ \partial_x\|\mathbf{y} - \mathbf{B}(\phi)\|_{T^{-1}} & \partial_y\|\mathbf{y} - \mathbf{B}(\phi)\|_{T^{-1}} \end{bmatrix}$$

We want to prove that:

$$det\left(\mathbb{J}_{(x,y)}\left(\mathbf{F}\right)\left(\phi,\mathbf{A}\left(\phi\right)\right)\right) = det\left(\mathbb{J}^{T}_{(x,y)}\left(\mathbf{F}\right)\left(\phi,\mathbf{A}\left(\phi\right)\right)\right) \neq 0.$$

This is true if and only if $\nabla_{(x,y)}\|\mathbf{y}\|_{T^{-1}}$ and $\nabla_{(x,y)}\|\mathbf{y}-\mathbf{B}\left(\phi\right)\|_{T^{-1}}$ are linearly independent.

In proposition 2.25, we proved that for a differentiable and strictly convex norm, $\nabla\|\mathbf{x}\|, \nabla\|\mathbf{y}\|$ define supporting lines in $\mathbf{x}\neq\mathbf{y}$, $\|\mathbf{x}\|_{T^{-1}} = \|\mathbf{y}\|_{T^{-1}} = 1$ respectively. Consider:

$$n_{\mathbf{x}} = \frac{\nabla\|\mathbf{x}\|_{T^{-1}}}{\|\nabla\|\mathbf{x}\|_{T^{-1}}\|_{T^{-1}}},\ n_{\mathbf{y}} = \frac{\nabla\|\mathbf{y}\|_{T^{-1}}}{\|\nabla\|\mathbf{y}\|_{T^{-1}}\|_{T^{-1}}}.$$

$\|n_{\mathbf{x}}\| = \|\lambda\cdot n_{\mathbf{y}}\| = |\lambda|\cdot\|n_{\mathbf{y}}\| \implies |\lambda| = 1.$

For $\lambda = 1$, $\underset{2.26}{\implies}$ $n_{\mathbf{x}}$ and $n_{\mathbf{y}}$ are not proportional.
For $\lambda = -1 : \nabla\|\cdot\|\big|_{\mathbf{x}} = -1\cdot\nabla\|\cdot\|\big|_{\mathbf{y}} = \nabla(\|\cdot\|)\big|_{-\mathbf{y}}$ due to the chain rule [1]
$\underset{2.26}{\implies}$ $\mathbf{y} = -\mathbf{x}$.
For $n_{\mathbf{y}}$, $n_{\mathbf{y}-\mathbf{B}(\phi)}$ the normalized vectors of $\nabla_{(x,y)}\|\mathbf{y}\|_{T^{-1}}$ and $\nabla_{(x,y)}\|\mathbf{y}-\mathbf{B}\left(\phi\right)\|_{T^{-1}}$ respectively:

$$n_{\mathbf{y}} = -n_{\mathbf{y}-\mathbf{B}(\phi)} \implies \mathbf{y}-\mathbf{B}\left(\phi\right) = -\mathbf{y}.$$

$$\implies \|2\cdot\mathbf{y}\|_{T^{-1}} = \|\mathbf{B}\left(\phi\right)\|_{T^{-1}} \implies \|\mathbf{y}\|_{T^{-1}} = \frac{1}{2}\lightning.$$

3.4.2 $\alpha(\phi)$ is injective

We now need to prove that $\alpha\left(\phi\right)$ is injective. Fix $\|\mathbf{A}\|_{T^{-1}} = \|\mathbf{B}\|_{T^{-1}} = 1$ s.t. $\|\mathbf{A}-\mathbf{B}\|_{T^{-1}} = \varepsilon$. We'll first consider a point $\mathbf{C}$ on the arc between $\mathbf{A}$ and $\mathbf{B}$.
Let $\mathbf{S} : [0,2\pi] \longrightarrow \mathbb{R}^2$ s.t. $\|\mathbf{S}\left(\phi\right)\|_{T^{-1}} = 1$. Define $\beta\left(\phi\right)$ as the angle between the ray $y \equiv \mathbf{C}_y$ starting in $\mathbf{C}$ and $\mathbf{B}-\mathbf{C} = \mathbf{v}\left(\beta(\phi)\right)$.
Let $\mathbf{A} = S(\phi_1)$, $\mathbf{B} = S(\phi_2)$ s.t. $\mathbf{B}-\mathbf{A} = \mathbf{v}\left(\alpha(\phi_1)\right) = \mathbf{v}\left(\beta(\phi_1)\right)$. Choose $\mathbf{C}\neq\mathbf{A}$.

We have that β is and increasing function, due to the strict convexity of the unit ball. By contradiction, assume there exist $\phi_3 \in (\phi_1,\phi_2)$ s.t. $\beta\left(\phi_3\right) = \alpha\left(\phi_1\right)$.
$\implies \mathbf{B}-\mathbf{C} = \mathbf{v}\left(\beta\left(\phi_3\right)\right) = c\cdot\mathbf{v}\left(\alpha(\phi_1)\right)$ (note that $c \lneqq 1$ as $\|\mathbf{B}-\mathbf{C}\|_{T^{-1}} \lneqq \varepsilon$).
Then $\mathbf{C} = \theta\cdot\mathbf{A} + (1-\theta)\cdot\mathbf{B}$ for some $\theta\in(0,1)$. Strict convexity implies that $\|C\| \lneqq 1\ \lightning$. $\implies \beta(\phi_3) \gneqq \beta(\phi_1) = \alpha(\phi_1)$ for $\phi_3 \gneqq \phi_1$.

Now consider a point $S(\bar{\phi})$ s.t. $\bar{\phi} \geqslant \phi_3$. Let $\gamma(\bar{\phi})$ be the angle between the ray $y \equiv \mathbf{C}_y$ starting in $\mathbf{C}$ and $S(\bar{\phi})-\mathbf{C}$. Now let $\mathbf{D} = \mathbf{S}(\phi_4)$ s.t. $\|\mathbf{D}-\mathbf{C}\|_{T^{-1}} = \varepsilon$. As proved before, we have that γ is non decreasing and $\gamma(\phi_3)\neq\gamma(\phi_4)$.
$\beta(\phi_3) = \gamma(\phi_3) \lneqq \gamma(\phi_4) = \alpha(\phi_3) \implies \alpha(\phi_3) \gneqq \beta(\phi_3) \gneqq \alpha(\phi_1)$. We have hereby proved that α is a monotonic increasing function, therefore it's injective.

[1] Note that $\mathbf{y}\neq 0 \implies \nabla\|\mathbf{y}\|\neq 0$.

3.4.3 The inequality

Let $\Delta(\varepsilon,\phi)$ be the norm of the segment joining the midpoint of $\overline{AB}$ and the unit sphere. In polar coordinates: let $\phi \in [0,2\pi]$, $r \in [\,0,\infty)$ such that $x = r\cdot\cos(\phi)$ & $y = r\cdot\sin(\phi)$.
Then

$$\mathbf{Y}_\varepsilon = \int_0^{2\pi}\int_0^{[1-\Delta(\phi,\varepsilon)]\cdot r(\phi)} r\,dr\,d\phi = \int_0^{2\pi} \frac{[1-\Delta(\phi,\varepsilon)]^2\cdot r^2(\phi)}{2}\,d\phi$$

$$\underset{(3.12)}{=} \left(1-\frac{\varepsilon^2}{4}\right)\int_0^{2\pi}\frac{r^2(\phi)}{2}\,d\phi$$

$$\Longrightarrow \int_0^{2\pi}\left[[1-\Delta(\varepsilon,\phi)]^2-\left(1-\frac{\varepsilon^2}{4}\right)\right]\frac{r^2(\phi)}{2}\,d\phi = 0$$

$$^{2}\Longrightarrow \inf_{\phi\in[0,2\pi]}(1-\Delta(\phi,\varepsilon))^2 \leqslant \left(1-\frac{\varepsilon^2}{4}\right)$$

$$\Longrightarrow \delta(\varepsilon) = \underset{\phi\in[0,2\pi]}{inf}\ \Delta(\phi,\varepsilon) \leqslant 1-\sqrt{1-\frac{\varepsilon^2}{4}}.$$

---

[2] $\underset{\phi\in[0,2\pi]}{inf}\ (1-\Delta(\phi,\varepsilon))^2 > \left(1-\frac{\varepsilon^2}{4}\right) \Longrightarrow \int_0^{2\pi}\left[[1-\Delta(\varepsilon,\phi)]^2-\left(1-\frac{\varepsilon^2}{4}\right)\right]\cdot r^2(\phi)\,d\phi > 0$

Chapter 4

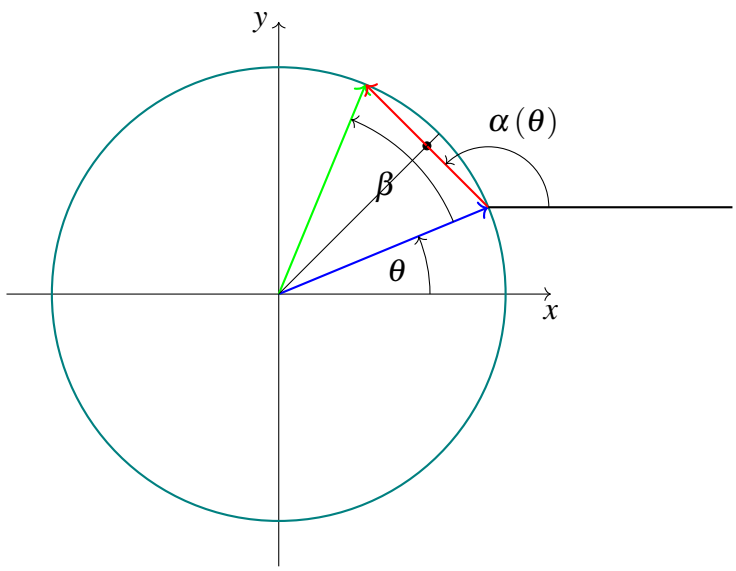
Nordlander's Theorem for l_2^2

In this section we will give a geometric interpretation to Nordlander's Theorem. We are going to explicitly calculate the modulus of convexity for the 2-Dimensional Euclidean space l_2^2.

Recall α the angle defined in 3.4. Here we'll give an explicit formula.

4.1 Calculating α

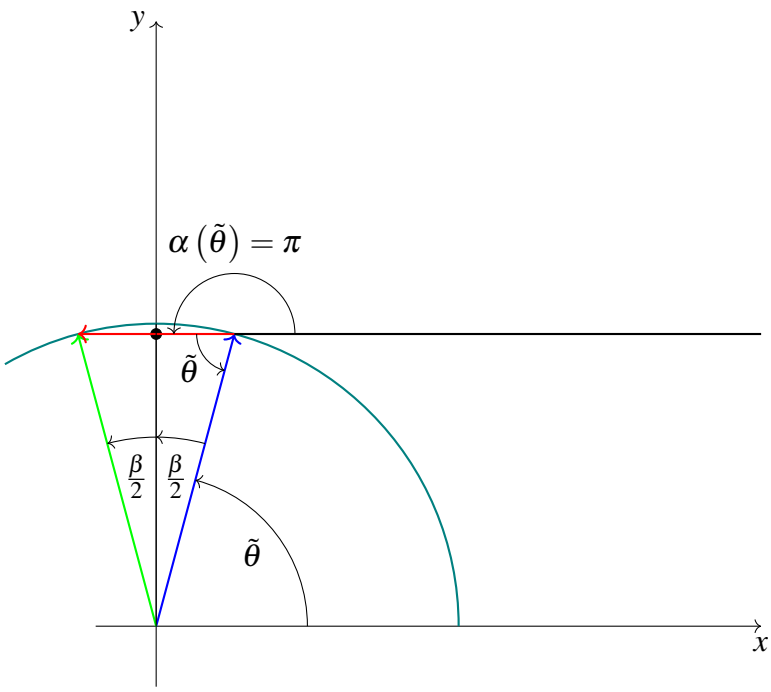
Given two point $\mathbf{A}, \mathbf{B}$ on the unit sphere in l_2^2 such that $\|A - B\|_2 = \varepsilon$ and $\mathbf{O} = (0,0)$; let β be the angle between $\overline{OA}$ and $\overline{OB}$ (*note* that β is fixed.).



Let $\alpha(\phi)$ be the angle between the ray $y \equiv \mathbf{A}_y$ starting in $\mathbf{A}(\phi)$ parallel to the x-axis and $\mathbf{B}(\phi) - \mathbf{A}(\phi)$. For simplicity, we'll find β for $\alpha(\theta) = \pi$.

Let $\tilde{\theta} \in \left[0, \frac{\pi}{2}\right]$ such that $\alpha(\tilde{\theta}) = \pi$.
Note that $\alpha(\theta) = \pi$ when the y-axis intersects the segment $\overline{AB}$ in it's midpoint. This is due the symmetrical nature of the unit sphere in l_2^2. The triangle formed by $\overline{AB}$,$\overline{OA}$,$\overline{OB}$ is isosceles, due to $\overline{OA}$ & $\overline{OB}$ being the radii of the circle.
With basic trigonometry, we can conclude:

$$\cos(\tilde{\theta}) = \frac{\varepsilon}{2} \text{ \& } \sin(\tilde{\theta}) = \sqrt{1 - \cos^2(\tilde{\theta})} = \sqrt{1 - \frac{\varepsilon^2}{4}}.$$



This also allows us to calculate:

$$\sin\left(\frac{\beta}{2}\right) = \frac{\varepsilon}{2} \text{ \& } \cos\left(\frac{\beta}{2}\right) = \sqrt{1 - \frac{\varepsilon^2}{4}}.$$

Therefore

$$\sin(\beta) = 2 \cdot \sin\left(\frac{\beta}{2}\right) \cdot \cos\left(\frac{\beta}{2}\right) = \varepsilon\sqrt{1 - \frac{\varepsilon^2}{4}} \tag{4.1}$$

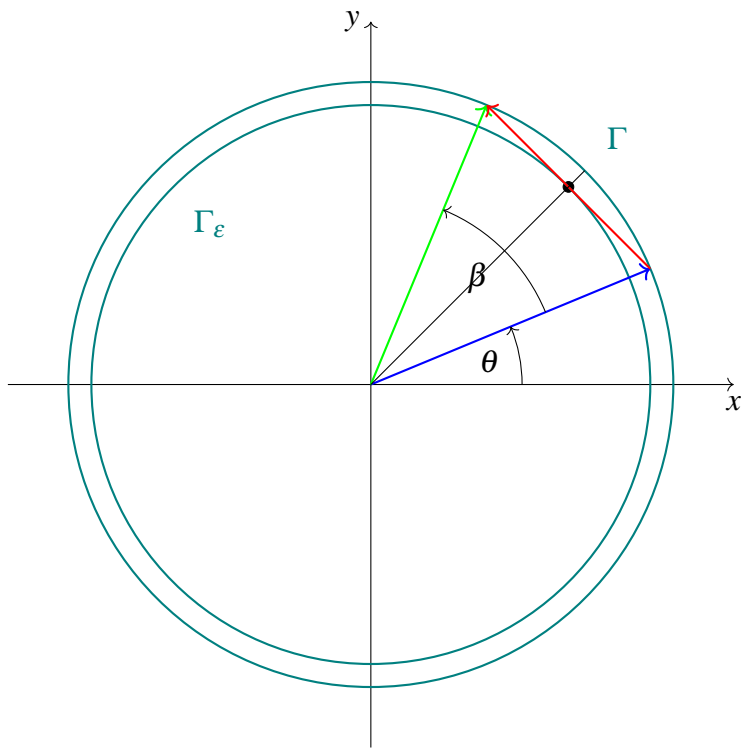
$$\cos(\beta) = \cos^2\left(\frac{\beta}{2}\right) - \sin^2\left(\frac{\beta}{2}\right) = 1 - \frac{\varepsilon^2}{2}. \tag{4.2}$$

We have also found an explicit formula for $\alpha(\theta) = \theta + \pi - \arccos\left(\frac{\varepsilon}{2}\right)$.[1]

[1] $\tilde{\theta} = \arccos\left(\frac{\varepsilon}{2}\right)$

4.2 Area within Γ_ε

Let Y be the area enclosed within $S^1 = \Gamma = \{\mathbf{y} \in \mathbb{R}^2 : \|\mathbf{y}\|_2 = 1\}$, and let Y_ε be the area enclosed within $\Gamma_\varepsilon = \left\{\frac{\mathbf{x}+\mathbf{y}}{2} : \mathbf{x}, \mathbf{y} \in S^1,\ \|\mathbf{x}-\mathbf{y}\|_2 = \varepsilon\right\}$. $Y = \pi$ as Γ is the unit circumference.



Recall in 3.4 we defined an integral $\mathbf{X}$. Here we'll give an explicit calculation of said integral.

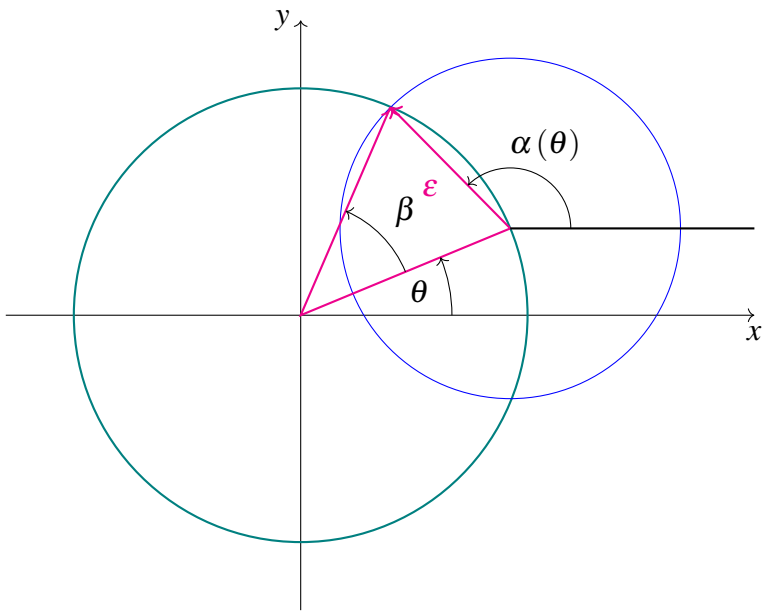
In the previous chapter we obtained:

$$\mathbf{A}(0) = \begin{pmatrix} \cos(\beta) \\ \sin(\beta) \end{pmatrix} = \begin{pmatrix} 1 - \frac{\varepsilon^2}{2} \\ \varepsilon\sqrt{1 - \frac{\varepsilon^2}{4}} \end{pmatrix}$$

$\mathbf{A}(\theta)$ can be written as an anti-clockwise rotation of

$$\implies \mathbf{A}(\theta) = \begin{pmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{pmatrix} \begin{pmatrix} 1 - \frac{\varepsilon^2}{2} \\ \varepsilon\sqrt{1 - \frac{\varepsilon^2}{4}} \end{pmatrix}.$$

Lets consider $\mathbf{v}(\theta)$ as the vector with end point $\mathbf{B}(\theta)$ and start point $\mathbf{A}(\theta)$ $\forall\theta \in [0, 2\pi]$. Our aim is to calculate the area $\mathbf{X}$ enclosed within the curve γ drawn by the end point of $\mathbf{v}(\alpha(\theta))$ with start point in the origin. We need to prove that the parametrization is injective and differentiable. This is true given that $\theta + c$ where $c \in \mathbb{R}$ is a injective and differentiable function. We will give a different proof later on using the implicit function theorem.



$$\text{First note that } \mathbf{v}(\alpha(\theta)) = \begin{pmatrix} \varepsilon \cdot \cos(\alpha(\theta)) \\ \varepsilon \cdot \sin(\alpha(\theta)) \end{pmatrix} = \begin{pmatrix} \varepsilon \cdot \cos(\theta + \pi - \tilde{\theta}) \\ \varepsilon \cdot \sin(\theta + \pi - \tilde{\theta}) \end{pmatrix}$$

$$= \begin{pmatrix} \varepsilon \cdot (-\cos(\theta - \tilde{\theta})) \\ \varepsilon \cdot (-\sin(\theta - \tilde{\theta})) \end{pmatrix} = \varepsilon \begin{pmatrix} -\cos(\theta)\cos(\tilde{\theta}) - \sin(\theta)\sin(\tilde{\theta}) \\ \sin(\tilde{\theta})\cos(\theta) - \sin(\theta)\cos(\tilde{\theta}) \end{pmatrix}$$

$$\text{Therefore } \mathbf{v}(\alpha(\theta)) = \begin{pmatrix} -\cos(\theta) & -\sin(\theta) \\ -\sin(\theta) & \cos(\theta) \end{pmatrix} \begin{pmatrix} \varepsilon \cdot \frac{\varepsilon}{2} \\ \varepsilon\sqrt{1 - \frac{\varepsilon^2}{4}} \end{pmatrix}.$$

$$\text{Also note that } \mathbf{v}(\theta) = \mathbf{B}(\theta) - \mathbf{A}(\theta) = \begin{pmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{pmatrix} \begin{pmatrix} -\frac{\varepsilon^2}{2} \\ \varepsilon\sqrt{1 - \frac{\varepsilon^2}{4}} \end{pmatrix}.$$

We have that:

$$v_x(\theta) = -\frac{\varepsilon^2}{2}\cos(\theta) - \varepsilon\sqrt{1-\frac{\varepsilon^2}{4}}\sin(\theta) \tag{4.3}$$

$$v_y(\theta) = \frac{\varepsilon^2}{2}\sin(\theta) + \varepsilon\sqrt{1-\frac{\varepsilon^2}{4}}\sin(\theta) \tag{4.4}$$

Applying Green's Theorem:

$$\mathbf{X} = \tfrac{1}{2}\oint_\gamma x\,dy - y\,dx = \tfrac{1}{2}\int_0^{2\pi}\left[v_y(\theta)\,\partial_\theta v_x(\theta) - v_x(\theta)\,\partial_\theta v_y(\theta)\right]d\theta \tag{4.5}$$

$$= \pi\left(\left(-\frac{\varepsilon^2}{2}\right)^2 + \varepsilon^2\left(\sqrt{1-\frac{\varepsilon^2}{4}}\right)^2\right) = \pi\cdot\varepsilon^2.$$

An easier way to calculate the area enclosed within Γ_ε: calculate the area of a circle of radius $r = \sin(\tilde{\theta}) = \sqrt{1-\frac{\varepsilon^2}{4}}$.
Therefore

$$\mathbf{Y}_\varepsilon = \pi\cdot r^2 = \pi\cdot\left(1-\frac{\varepsilon^2}{4}\right) = \left(1-\frac{\varepsilon^2}{4}\right)\mathbf{Y}. \tag{4.6}$$

4.3 Another proof of the differentiability of $\alpha(\theta)$

Here we provide another proof that $\alpha(\theta)$ is differentiable.
We want to apply the implicit function theorem to prove that $\mathbf{A}(\theta)$ is differentiable, implying that $\alpha(\theta) = \arctan\left(\frac{\mathbf{v}_y(\theta)}{\mathbf{v}_x(\theta)}\right) = \arctan\left(\frac{\mathbf{A}_y(\theta)-\sin(\theta)}{\mathbf{A}_x(\theta)-\cos(\theta)}\right)$ is differentiable. Lets consider the function $\mathbf{F} : [0,2\pi] \times \mathbb{R}^2 \to \mathbb{R}^2$ such that:

$$\mathbf{F}(\theta,\mathbf{y}) = [\|\mathbf{y}\|_2 - 1, \|\mathbf{y}-\mathbf{x}(\theta)\|_2 - \varepsilon] \tag{4.7}$$

where $\mathbf{y} \in \mathbb{R}^2$ and $\mathbf{x}(\theta) = \mathbf{B}(\theta)$ is a point on the unit ball.

$$\mathbf{F}(\theta,\mathbf{A}(\theta)) \equiv \mathbf{0}\ \forall\theta \in [0,2\pi]. \tag{4.8}$$

$$\mathbb{J}_{(\theta,\mathbf{y})}(\mathbf{F}) = \left[\begin{array}{c|cc} \frac{\partial F_1}{\partial\theta} & \frac{\partial F_1}{\partial x} & \frac{\partial F_1}{\partial y} \\ \frac{\partial F_2}{\partial\theta} & \frac{\partial F_2}{\partial x} & \frac{\partial F_2}{\partial y} \end{array}\right] = \left[\mathbb{J}_\theta(\mathbf{F}) \,\middle|\, \mathbb{J}_{(x,y)}(\mathbf{F})\right]$$

To prove that $\mathbf{A}(\theta)$ is differentiable, we must first prove that $\mathbb{J}_{(x,y)}(\mathbf{F})(\theta,\mathbf{A}(\theta))$ is invertible:

$$\mathbb{J}_\theta(\mathbf{A})(\theta) = -\mathbb{J}^{-1}_{(x,y)}(\mathbf{F})(\theta,\mathbf{A}(\theta)) \cdot \mathbb{J}_\theta(\mathbf{F})(\theta,\mathbf{A}(\theta)).$$

$$\mathbb{J}_{(x,y)}(\mathbf{F}) = \begin{bmatrix} \partial_x\|\mathbf{y}\|_2 & \partial_x\|\mathbf{y}\|_2 \\ \partial_x\|\mathbf{y}-\mathbf{x}(\theta)\|_2 & \partial_x\|\mathbf{y}-\mathbf{x}(\theta)\|_2 \end{bmatrix} = \begin{bmatrix} \frac{x}{\|\mathbf{y}\|_2} & \frac{y}{\|\mathbf{y}\|_2} \\ \frac{x-\cos(\theta)}{\|\mathbf{y}-\mathbf{x}(\theta)\|_2} & \frac{y-\sin(\theta)}{\|\mathbf{y}-\mathbf{x}(\theta)\|_2} \end{bmatrix}.$$

We want to prove that:

$$det\left(\mathbb{J}_{(x,y)}(\mathbf{F})(\theta,\mathbf{A}(\theta))\right) = det\left(\mathbb{J}^T_{(x,y)}(\mathbf{F})(\theta,\mathbf{A}(\theta))\right) \neq 0.$$

This is true if and only if $\nabla_{(x,y)}\|\mathbf{y}\|_2 = \frac{\mathbf{y}}{\|\mathbf{y}\|_2}$ and $\nabla_{(x,y)}\|\mathbf{y}-\mathbf{x}(\theta)\|_2 = \frac{\mathbf{y}-\mathbf{x}(\theta)}{\|\mathbf{y}-\mathbf{x}(\theta)\|_2}$ are linearly independent. Note that $\|\nabla_{(x,y)}\|\mathbf{y}\|_2\|_2 = \|\nabla_{(x,y)}\|\mathbf{y}-\mathbf{x}(\theta)\|_2\|_2 = 1$.

If $\mathbf{v},\mathbf{w} \in \{\|\mathbf{y}\|_2 = 1 : \mathbf{y} \in \mathbb{R}^2\}$ such that $\mathbf{v} = \lambda\mathbf{w}$, then $|\lambda| = 1$

If $\nabla_{(x,y)}\|\mathbf{y}\|_2 = \mathbf{y}$ & $\nabla_{(x,y)}\|\mathbf{y}-\mathbf{x}(\theta)\|_2 = \dfrac{\mathbf{y}-\mathbf{x}(\theta)}{\varepsilon}$ are proportional,

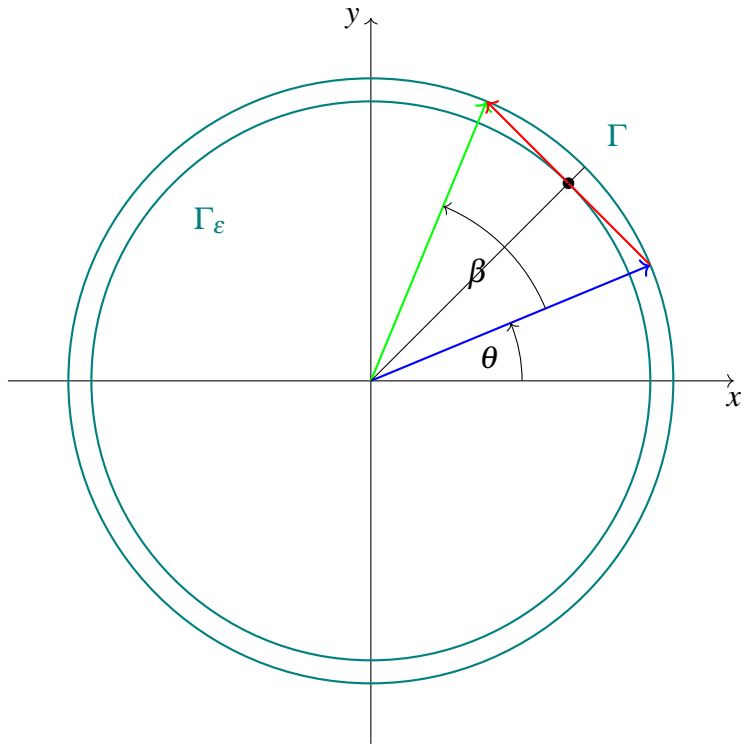
then

$$\nabla_{(x,y)}\|\mathbf{y}\|_2 = \nabla_{(x,y)}\|\mathbf{y}-\mathbf{x}(\theta)\|_2.$$

However $\mathbf{y}$ and $\mathbf{y}-\mathbf{x}(\theta)$ cannot be proportional unless $\mathbf{x}(\theta)$ is proportional to $\mathbf{y}$ itself. With the Euclidean norm it's easy to see that this is not the case for $\varepsilon \in (0,2)$.

4.4 The Modulus of convexity

Let $\Delta(\varepsilon,\theta)$ be the distance between the midpoint of $\overline{AB}$ and the unit sphere. In polar coordinates: let $\theta \in [0,2\pi], r \in [\,0,\infty)$ such that $x = r\cdot\cos(\theta)$ & $y = r\cdot\sin(\theta)$.



Then:

$$\mathbf{Y}_\varepsilon = \int_0^{2\pi}\int_0^{1-\Delta(\theta,\varepsilon)} r\,dr\,d\theta = \int_0^{2\pi} \frac{[1-\Delta(\theta,\varepsilon)]^2}{2}\,d\theta \underset{(4.6)}{=} \left(1-\frac{\varepsilon^2}{4}\right)\int_0^{2\pi}\frac{1}{2}\,d\theta$$

$$\Longrightarrow \int_0^{2\pi}\left[[1-\Delta(\varepsilon,\theta)]^2-\left(1-\frac{\varepsilon^2}{4}\right)\right]d\theta = 0\,^2 \Longrightarrow \inf_{\theta\in[0,2\pi]}(1-\Delta(\theta,\varepsilon))^2 \leqslant \left(1-\frac{\varepsilon^2}{4}\right)$$

$$\Longrightarrow \delta(\theta) = \inf_{\theta\in[0,2\pi]}\Delta(\theta,\varepsilon) \leqslant 1-\sqrt{1-\frac{\varepsilon^2}{4}}.$$

[2] $\inf_{\theta\in[0,2\pi]}(1-\Delta(\theta,\varepsilon))^2 > \left(1-\frac{\varepsilon^2}{4}\right) \Longrightarrow \int_0^{2\pi}\left[[1-\Delta(\varepsilon,\theta)]^2-\left(1-\frac{\varepsilon^2}{4}\right)\right]d\theta > 0$

For the unit sphere in l_2^2 :

$$\delta(\varepsilon) = \inf_{\theta \in [0,2\pi]} \Delta(\theta, \varepsilon) \geqslant \Delta\left(\bar{\theta}, \varepsilon\right) - \varepsilon' = {}^{3}1 - \sqrt{1 - \frac{\varepsilon^2}{4}} - \varepsilon'$$

for $\bar{\theta} \in [0, 2\pi], \varepsilon' > 0$.
ε' is arbitrary: therefore $\delta(\varepsilon) \geqslant 1 - \sqrt{1 - \frac{\varepsilon^2}{4}}$.

[3] $\Delta(\theta, \varepsilon) = 1 - \sqrt{1 - \frac{\varepsilon^2}{4}}\ \forall \theta \in [0, 2\pi]$

Bibliography

[Cla36] James A. Clarkson. Uniformly convex spaces. *Transactions of the American Mathematical Society*, 40(3):396–414, 1936.

[Fol99] Gerald B. Folland. *Real Analysis: Modern Techniques and Their Applications*, volume 195. John Wiley & Sons, 2 edition, 1999.

[GH96] Mariano Giaquinta and Stefan Hildebrandt. *Calculus of Variations II: The Hamiltonian Formalism*, volume 311 of *Grundlehren der mathematischen Wissenschaften*. Springer-Verlag, 1996.

[Nor60] Göte Nordlander. The modulus of convexity in normed linear spaces. *Arkiv för Matematik*, 4(1):15–17, 1960.